

A Systematic Exploration of Text Decomposition and Budget Distribution in Differentially Private Text Obfuscation

Stephen Meisenbacher, Angelo Kleinert, and Florian Matthes

Technical University of Munich

School of Computation, Information and Technology

Department of Computer Science

Garching, Germany

{stephen.meisenbacher, angelo.kleinert, matthes}@tum.de

Abstract

The goal of *differentially private text obfuscation* is to obfuscate, or “perturb”, input texts with Differential Privacy (DP) guarantees, such that the private output texts are quantifiably indistinguishable from the originals. While perturbation at the word level is intuitive, meaningful text privatization happens on complete documents. Recent research has laid the groundwork for reasoning about *privacy budget distribution*, namely, how an overall ϵ budget can be sensibly distributed among the component pieces of a text. We perform a systematic evaluation of multiple text decomposition and budget distribution techniques in the context of DP text obfuscation, testing how different methods for chunking texts can be combined with techniques for allocating ϵ to these chunks. Our experiments reveal that such design choices are very important, as even with comparable privacy budgets, significantly different results can occur based on which methods are chosen. In this, we provide credible evidence of the feasibility of maximizing empirical trade-offs by optimizing DP obfuscation procedures.

1 Introduction

In the rapidly advancing world of AI and LLMs, the reliance of the modern technological economy on massive data collection has justifiably raised concerns of privacy (Yao et al., 2024). Calls for privacy protection have been met by a significant string of privacy research in the context of Natural Language Processing (NLP) (Pan et al., 2020; Mahendran et al., 2021; Sousa and Kern, 2023). One theoretically viable, but practically challenging solution comes in the form of text privatization under Differential Privacy (DP) guarantees (Igamberdiev et al., 2022), thus spring-boarding numerous works at the intersection of DP and NLP (Hu et al., 2024).

One of the immediate challenges of privatizing text under DP comes with reasoning about the *unit of privatization* (Klymenko et al., 2022). Many of

the solutions to this challenge propose DP text obfuscation at the sub-document level, such as with the perturbation of words (Feyisetan et al., 2020) or tokens (Mattern et al., 2022), thereby creating a new challenge of meaningful text privatization for downstream usage. Relying on the compositionality of DP (Dwork, 2006) – whereby repeated perturbations on the same (text) data are *composed* (Feyisetan et al., 2020) – recent works have begun to conduct evaluations of sub-document-level perturbation mechanisms on full documents (Meisenbacher et al., 2024). This is achieved by assigning a privacy budget (the ϵ parameter of DP) to the document, and subsequently allocating this budget among component pieces (e.g., words).

While a simple approach may be to evenly distribute a privacy budget to all component words of a document (top of Figure 1), recent work has proposed techniques for more intelligently performing this allocation (bottom of Figure 1) (Meisenbacher et al., 2025). Similarly, other work demonstrates the value in more complex *text decomposition* methods, where an input document is *chunked* dynamically, for example by phrases and n-grams (Kim et al., 2021; Meisenbacher et al., 2024), or even sentences (Meehan et al., 2022). The combination of these techniques, i.e., dynamic text decomposition *and* privacy budget distribution, has neither been explored nor systematically evaluated.

We design a systematic evaluation of text decomposition and privacy budget distribution with DP text obfuscation, asking the following questions:

- RQ1. How can methods for text decomposition and privacy budget allocation be combined and evaluated for DP text obfuscation?
- RQ2. Are there significant differences, particularly in the privacy-utility trade-offs, between the combinations of RQ1?

Designing an experimental setup with two datasets, three privatization levels (ϵ), five decom-

Naive:	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	= 14
	Life	is	like	riding	a	bicycle.	To	keep	your	balance,	you	must	keep	moving	
Distributed:	0.37	0.56	1.48	0.94	2.01	0.57	1.98	0.84	1.34	0.65	1.32	0.67	0.86	0.40	= 14
	Life	is	like	riding	a	bicycle.	To	keep	your	balance,	you	must	keep	moving	

Figure 1: An example of text decomposition and budget distribution for DP text obfuscation. Given the same input text of 14 words (excluding punctuation), one can *decompose* the text into meaningful chunks beyond simple word tokenization. In addition, with a privacy budget of 14, one can either “naively” (uniformly) allocate it, or rather, distribute the budget sensibly such that more important words receive higher privatization (lower ϵ), and vice versa. Note: for understandability, we illustrate word-level budgets; we simply add these for the chunk-level budget.

position methods and six distribution methods, we comparatively analyze **180** different DP text obfuscation configurations, evaluating these setups on privacy, utility, and the trade-offs between the two.

While we find that there does not exist a universally dominant combination of methods in terms of privacy *and* utility, our results demonstrate that privacy, utility, *or* trade-offs can be optimized depending on which decomposition-distribution setup is chosen. These findings are supported by significant differences in metrics between setups, showcasing that such design choices *do* matter and have quantifiable and practically relevant implications.

Our work advances the field of DP text obfuscation by comprehensively exploring different avenues for transforming local DP perturbations into usable and optimized privatized documents. In particular, we make the following contributions:

1. We systematically evaluate 180 setups for DP text obfuscation, under various text decomposition and budget distribution schemes.
2. We perform a comparative analysis of privacy and utility preservation, demonstrating significant differences between setups.
3. We open-source our modular code at <https://github.com/sjmeis/DP-Decompose-Distribute>.

2 Foundations and Related Work

DP and Text Privatization. DP (Dwork, 2006) ensures a level of formal privacy by bounding the contribution of any individual in a dataset to queries or computations performed on the data. This is governed by the privacy parameter (ϵ , or the *privacy budget*), which affords *indistinguishability* to the individual. Formally, this is expressed as:

$$Pr[\mathcal{M}(D) \in \mathcal{S}] \leq e^\epsilon Pr[\mathcal{M}(D') \in \mathcal{S}],$$

for any databases D and D' differing in exactly one element (or “individual”), any $\epsilon > 0$, any function $\mathcal{M}$, and all $\mathcal{S} \subseteq \text{Range}(\mathcal{M})$.

In the context of text privatization, where DP is applied directly on the data itself, the notion of *local* DP (Kasiviswanathan et al., 2008) is often adopted. In this paradigm, the task of achieving DP is shifted to the user level, instead of some central curator. This, however, imposes a stricter indistinguishability requirement, as applying local DP considers the entire *universe* of data values rather than those contained within the dataset D of central (global) DP. As such, the DP notion becomes:

$$Pr[\mathcal{M}(x) = z] \leq e^\epsilon Pr[\mathcal{M}(x') = z]$$

Thus, an observed output cannot be attributed to *any* data point within a bounded probability.

The local DP paradigm is useful for applications in text privatization, as it allows for local perturbations of text representations, where individual data points can be likened to units of language, such as words or documents (Klymenko et al., 2022).

Metric Local DP. Some of the first proposed solutions to bring DP to the task of text privatization leveraged a generalized notion called *metric* (local) DP (MLDP) (Chatzikokolakis et al., 2013). MLDP relaxes the strict requirement of local DP, namely the requirement of indistinguishability between *any* two data points, by scaling the privacy loss based on proximity in some metric space (formally, by scaling ϵ by some distance metric d). Early works in DP text privatization recognized the adaptability of MLDP to embedding spaces (Fernandes et al., 2019; Feyisetan et al., 2020), e.g., by *obfuscating* words via their embedding representations.

Applications of MLDP to text take one of two forms: direct obfuscation of embedding representations (Feyisetan et al., 2020) or modeling text replacement as a selection problem (Yue et al., 2021). We focus on the former, which views obfuscation holistically and does not constrain replacement candidates to a finite set. In particular, MLDP perturbations typically operate in three

steps (De Faveri et al., 2025): (1) *embedding*, (2) *perturbation* via DP noise, and (3) *projection* to a replacement (word). While most DP text obfuscation mechanisms follow this general paradigm, MLDP allows for flexibility in the mechanism’s design, resulting in numerous works that improve the usability of DP outputs (Carvalho et al., 2023; Arnold et al., 2023a,b; De Faveri et al., 2025). Extending beyond word-level perturbations, Feyisetan et al. (2020) provide a framework for *document-level* obfuscation via basic DP composition.

Particularly in the evaluation of MLDP text obfuscation approaches, Meisenbacher et al. (2025) highlight an important consideration with word-level approaches, namely the need to test with uniform document-level privacy budgets; otherwise, comparability is lacking if these budgets are not equal (i.e., when composition of word-level budgets is unbounded). We ground our work in this, as we seek to optimize the allocation of a fixed privacy budget among the components of a text.

Text Chunking and Multi-Word Expressions.

The task of text chunking is an early NLP problem that aims to segment texts into coherent chunks (Beeferman et al., 1999). While word segmentation is the simplest approach, other techniques logically separate phrases or sentences (Pak and Teh, 2017). Pecina and Schlesinger (2006) survey 82 association measures for extracting *collocations*, or meaningful groupings of words that often appear together. More generally, multi-word expressions (MWEs) (Sag et al., 2002) have been widely studied (Constant et al., 2017), pointing to the importance of linguistic units beyond word boundaries.

Our work draws motivation from DP text privatization research operating between the word and document level, such as with collocations (Meisenbacher et al., 2024) or sentences (Meehan et al., 2022). We focus on MWEs, as they strike a balance between words, which lack context, and sentences, which often are not cohesive units of expression.

3 Methodology

We outline the steps of our methodology, which span a preparation and privatization pipeline, culminating in an extensive evaluation. The workflow of our methodology is illustrated in Figure 2.

3.1 A note on privacy guarantees

We preface the introduction of text decomposition and privacy budget distribution methods with an

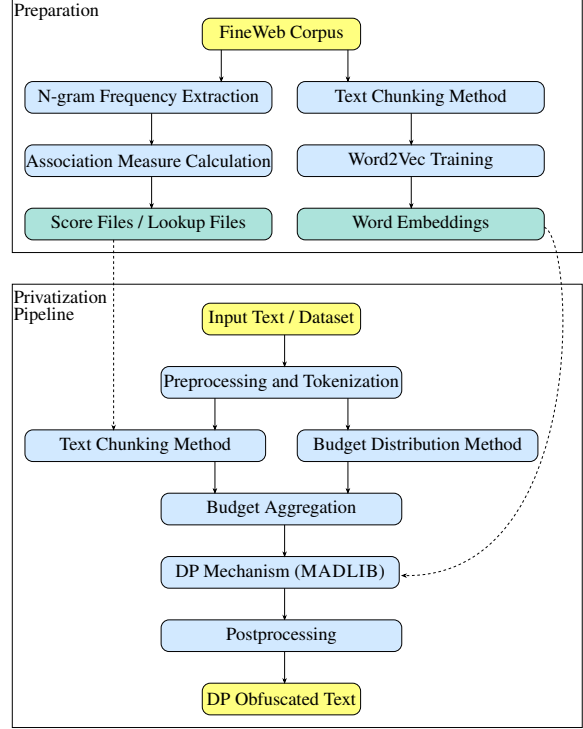


Figure 2: Workflow of our systematic evaluation.

important clarification on the *privacy guarantees* offered as a result of our privatization procedures. We measure and evaluate all texts on the **document level**; by leveraging the basic composition theorem of DP, we can *compose* the individual privatization of decomposed text chunks into a fixed, document-level privacy budget. This is essential to ensure that in evaluation, all texts receive the same document-level privacy budget, regardless of the number of decomposed units. The main goal of privacy budget distribution, therefore, is to allocate this overall budget to the component chunks, in an optimized manner. We detail both decomposition and budget distribution in the following.

3.2 N-gram Extraction

As three of our five text decomposition methods (Section 3.4) rely on association measures, the first step was to extract a corpus of n-grams for score calculation. We used the FineWeb dataset (Penedo et al., 2024), specifically the SAMPLE-10BT subset with 10B tokens. The goal of using this dataset was to extract commonly occurring English n-grams, to serve as the basis of text decomposition methods.

We extracted n-grams for $n \in \{1, 2, 3, 4\}$. To create a unified tokenization strategy for n-gram extraction (and for the remainder of this work), we used the simple regex `\b\w+\b`, using Python’s RE.

3.3 Association Measure Calculation

The next step was to quantify the strength of association between all n-grams (above unigrams), i.e., to detect the most meaningful groups of words. For this, we used three scoring techniques, following recommendations from related work (Bhalla and Klimcikova, 2019; Gu et al., 2021).

Pointwise Mutual Information (PMI). PMI measures how often words appear together compared to pure chance (Church and Hanks, 1990). We adopt a similar scheme to Meisenbacher et al. (2024) for PMI scoring, e.g., for bigram PMI:

$$\text{PMI}(w_1, w_2) = \log_2 \frac{c(w_1, w_2) \cdot N}{c(w_1) \cdot c(w_2)}$$

c represent a frequency count from the FineWeb corpus, and N is the total count. We adapt the PMI formula accordingly for trigrams and quad-grams, found in Appendix B. Following PMI calculation, we kept only n-grams with a frequency ≥ 275 and with a $\text{PMI} > 2$ (following related work), in order to mitigate bias towards lower frequency n-grams.

Log-Likelihood-Ratio (LLR). LLR is a statistical measure similar to PMI that compares observed frequencies of n-grams to their expected frequencies (Dunning, 1993). LLR is calculated from a contingency table of likelihood values, i.e., when two words appear together versus not. LLR is defined for bigrams but can be extended to trigrams and quad-grams (detailed in Appendix B). We kept the top 5% of LLR-scored n-grams (for $n > 1$).

t-score. The t-score also associates observed counts of n-grams with expected counts (Church and Hanks, 1990), normalized by the standard deviation. The bigram t-score is as follows:

$$t(w_1, w_2) = \frac{c(w_1, w_2) - \frac{c(w_1) \cdot c(w_2)}{N}}{\sqrt{c(w_1, w_2)}}$$

The adapted formulas for trigrams and quad-grams can be found in Appendix B. As with LLR, we kept the top 5% of t-scored n-grams.

3.4 Text Decomposition

We use five decomposition methods, which take a text as input and return a list of sequential MWEs.

Association-based Decomposition. Using each of the three selected association measures, we designed a greedy decomposition approach, which reads a text from left to right and selects the longest

available n-gram match (e.g., “all over the world” $\gg$ “all over the” $\gg$ “all over” $\gg$ “all”). This greedy approach was chosen over a score maximization approach, as the latter did not produce significantly different results in early testing. The complete algorithm for association-based decomposition is found in Algorithm 1 of the Appendix.

POS-based Decomposition. POS-based decomposition segments a text according to defined rules, for example, by combining noun or prepositional phrases. Rather than manually define rules, we trained a BIGRAMTAGGER from NLTK on the CoNLL 2000 shared task data (Tjong Kim Sang and Buchholz, 2000), which was focused on text chunking. With this tagger, an input text is decomposed into chunks based on the assigned POS tags, outlined in Algorithm 2 of the Appendix.

WordNet-based Decomposition. The final method uses WordNet lexical database (Miller, 1995) to identify matching n-grams that exist as synsets (entries). As with the associated measures, we read a text left to right and greedily select the longest available matching n-gram that exists in WordNet. For this implementation (Algorithm 3), we use WordNet from NLTK (Bird et al., 2009).

A Note on Stopwords and Contractions. To handle stopwords, we retain all stopwords *within* n-grams (i.e., the second word in a trigram, or second or third in a quad-gram). Otherwise, stopwords (from NLTK) are stripped from n-grams, and furthermore, are ignored during privatization.

Similarly, we opted to split contractions, thereby treating them as two words (“don” and “t”). To remedy this in the text reconstruction phase post-privatization, we recombine pairs of words that match a curated set of common contractions¹.

Embedding Model Training. To enable DP text obfuscation *chunk-wise*, an embedding model was trained for each of the five decomposition methods. In this way, a unified embedding model can be created, in which unigrams coexist with bigrams to quad-grams. For this, we train WORD2VEC models (Mikolov et al., 2013) using the Gensim² implementation with 300-d vectors and all default training values. N-grams are trained together with unigrams by treating n-grams as single words (e.g., “all_over_the_world”), thereby capturing relation-

¹<https://gist.github.com/J3RN/ed7b420a6e1d5bd6d06>

²<https://radimrehurek.com/gensim>

ships between n -grams regardless of n . The resulting models served as the basis for DP text obfuscation via embedding perturbation.

3.5 Privacy Budget Distribution

Privacy budget distribution assigns each chunk in a decomposed text a fraction of the overall budget (ϵ), such that the total budget is upheld and optimized. Pseudocode for some of the methods can be found in Algorithms 4-6 of the Appendix.

Attention Weights. We use attention weights from a BERT-BASE-UNCASED model (Devlin et al., 2019). After performing a forward pass of an input text through the model, we capture and average attention scores across all 12 attention heads, and again across all 12 layers. Finally, we average the resulting scores across the token dimension to obtain a single scalar score per input. Subword tokens were handled by summing the component scores. In addition, special tokens from BERT were filtered out. The final scores were normalized and distributed proportionally across all non-stopwords.

Integrated Gradients. We also use Integrated Gradients (Sundararajan et al., 2017) to capture attention. We first defined a forward function that maps input embeddings to a scalar score, which is simply the sum of the mean-pooled embedding (final hidden state) for each token. Then, using Captum³, we calculated attribution vectors for each input token, reaching a final score for budget distribution by using the L2 norm of the vectors. We then filtered out special tokens and normalized the scores, only distributing scores to non-stopwords.

Information Content. Information Content (IC) is a measure of the informational value of a unit of language, and it is calculated based on the frequency of occurrences in a corpus. We utilize pre-calculated IC values from NLTK, which contains five variants from different corpora. Since these IC values are tied to WordNet synsets, we first used the LESK function from NLTK to disambiguate word sense, and then calculated an IC score by averaging the entry values over all five IC corpora.

KEYBERT. KEYBERT (Grootendorst, 2020) is an unsupervised keyword extraction method leveraging BERT-like embedding models. We use KEYBERT with the ALL-MINILM-L6-V2 model (Reimers and Gurevych, 2019), to return a score

for all unigrams in an input text (i.e., $top-N$ for $N = len(text)$). Since a higher KEYBERT score indicates higher importance as a keyword, we invert these scores for budget distribution.

YAKE. We evaluate another unsupervised keyword extraction method, YAKE (Campos et al., 2020), which is statistical-based rather than transformer-based. As with KEYBERT, we ensure that YAKE returns a score for all unigrams. In contrast, though, lower scores indicate more suitable keyword candidates; thus, scores are left unaltered.

Final Budget Distribution. The output of each distribution method is a mapping of individual words from the input text to a corresponding score. To reach a final budget allocation, negative scores were handled by adding the absolute value of the minimum score to all scores. Next, stopword scores were set to 0. If necessary, the remaining scores were inverted. Finally, the scores were normalized to add up to 1, and then scaled by the target ϵ budget to fulfill the total budget constraint. Finally, scores belonging to the same decomposed chunk were summed. The steps taken to reach the final distribution and aggregate chunk scores are described in Algorithms 7 and 8, respectively.

4 Experimental Setup

We design a full factorial experiment, in which all decomposition methods are tested in combination with the selected budget distribution techniques. In addition to the five distribution techniques introduced above, we also test a “baseline” method, which is an even distribution of the overall ϵ (i.e., $\epsilon/len(words)$). Thus, a 5x6 factorial experiment is conducted, across two datasets and three privacy budgets, for a total of 180 experimental runs.

4.1 Datasets

We use two datasets of user-written texts, which present a plausible case for text obfuscation, where each dataset features a one-to-many mapping of authors to texts, for a constrained set of authors.

Trustpilot Reviews. We use a subset of the Trustpilot corpus (Hovy et al., 2015), a large corpus of user reviews. We take a 10k random sample of EN-US reviews, which are mapped to either negative sentiment (1-2 stars) or positive sentiment (4-5 stars); neutral reviews are not included. The reviews are also mapped to the gender of the author.

³<https://captum.ai/>

Privacy Level	Trustpilot	Yelp
High ($\epsilon = 0.1 \times \text{avg. doc length}$)	5.2	18.7
Medium ($\epsilon = 1 \times \text{avg. doc length}$)	52	187
Low ($\epsilon = 5 \times \text{avg. doc length}$)	260	935

Table 1: Document-level privacy budget (ϵ) values.

Yelp Reviews. Finally, we utilize a dataset of the 10 most-frequently writing authors on the Yelp platform, prepared by Utpala et al. (2023), taking a 10k random sample. As with Trustpilot, each review is also mapped to a binary sentiment score.

4.2 Privatization

Each privatization configuration (*decomposition, distribution, dataset*) is run on three *privacy levels*, represented by the document-level ϵ budget. As introduced, all texts within a dataset are obfuscated with the same ϵ level. The exact distribution of this budget among the component chunks of a text is determined by the distribution method in use.

To establish the ϵ budgets for the three privacy levels (*high, medium, low*), we choose base ϵ values, namely, $\epsilon \in \{0.1, 1, 5\}$, respectively. These values were then scaled by the average document length (in words) to achieve the final three document-level budgets per dataset (Table 1). With these, we ensure comparable privacy levels between all privatized texts of a dataset, while also scaling reasonably to the average document length.

Given an input text and a document-level ϵ , the text is first decomposed into chunks, and then each of these chunks is allocated a portion of the privacy budget via a distribution method. Finally, using the MADLIB method (Feyisetan et al., 2020), an MLDP perturbation mechanism, we obfuscate each of these chunks using the embedding model corresponding to the utilized decomposition method (Section 3.4). Each chunk is mapped to its embedding in our trained model, calibrated Laplacian noise according to the allocated budget is added to the embedding by MADLIB, and the perturbed embedding is projected back to the nearest embedding. The corresponding n-gram is then inserted as the DP obfuscated output. The output is postprocessed by removing “_” characters that connect n-grams and recombining contractions, where necessary.

4.3 Evaluation

Our evaluation takes the form of privacy and utility measurements, as well as the privacy-utility trade-off, which are detailed in the following.

4.3.1 Privacy Evaluation

The privacy evaluations analyze two aspects of privacy protection: personal identifier masking and defense against attribute inference.

To enumerate identifiers, we use Microsoft Presidio⁴ to detect all private identifiers (PI) in an input text. Then, we measure what percentage of these identifiers are still (completely) present in the obfuscated output texts. This is averaged over all original-private text pairs, where a lower average score represents better overall privacy protection.

To simulate attribute inference attacks, we adopt the adversarial framework of *static* and *adaptive* attackers (Mattern et al., 2022; Utpala et al., 2023). The static attacker is capable of training an adversarial classification on original, non-obfuscated “public” texts, thereafter using the trained model to infer sensitive attributes of DP obfuscated texts, i.e., the identity or gender of the original author. The adaptive attacker is further capable of mimicking the DP obfuscation process by first privatizing the training set, training the adversarial model on this data, and then inferring attributes of the target obfuscated texts. For both setups, we train a DEBERTA-V3-BASE model (He et al., 2021) for one epoch and use default Hugging Face Trainer parameters, except for the use of the focal loss function, a learning rate of 2e-5, and 500 warmup steps. The training set is a 90% random split of each dataset, and the target split is the remaining 10% test set. The resulting score is represented by the adversarial F1 inference performance.

4.3.2 Utility Evaluation

The utility evaluations capture three aspects: downstream utility, semantic similarity, and coherence.

For downstream utility, we fine-tune a DEBERTA-V3-BASE model for one epoch on a 90% train split of all Trustpilot and Yelp datasets, using the same parameters as the attacker models described above. The F1 score of the trained model on the 10% test set is reported as an average of three training runs.

To measure semantic similarity between original and obfuscated text counterparts, we measure the average cosine similarity between all pairs in a given dataset. This is performed by using the embeddings calculated by three different pre-trained sentence transformer models, namely ALL-MINI-LM-L6-V2 (Reimers and Gurevych, 2019), ALL-MPNET-BASE-V2 (ibid), and GTE-SMALL (Li

⁴<https://microsoft.github.io/presidio/>

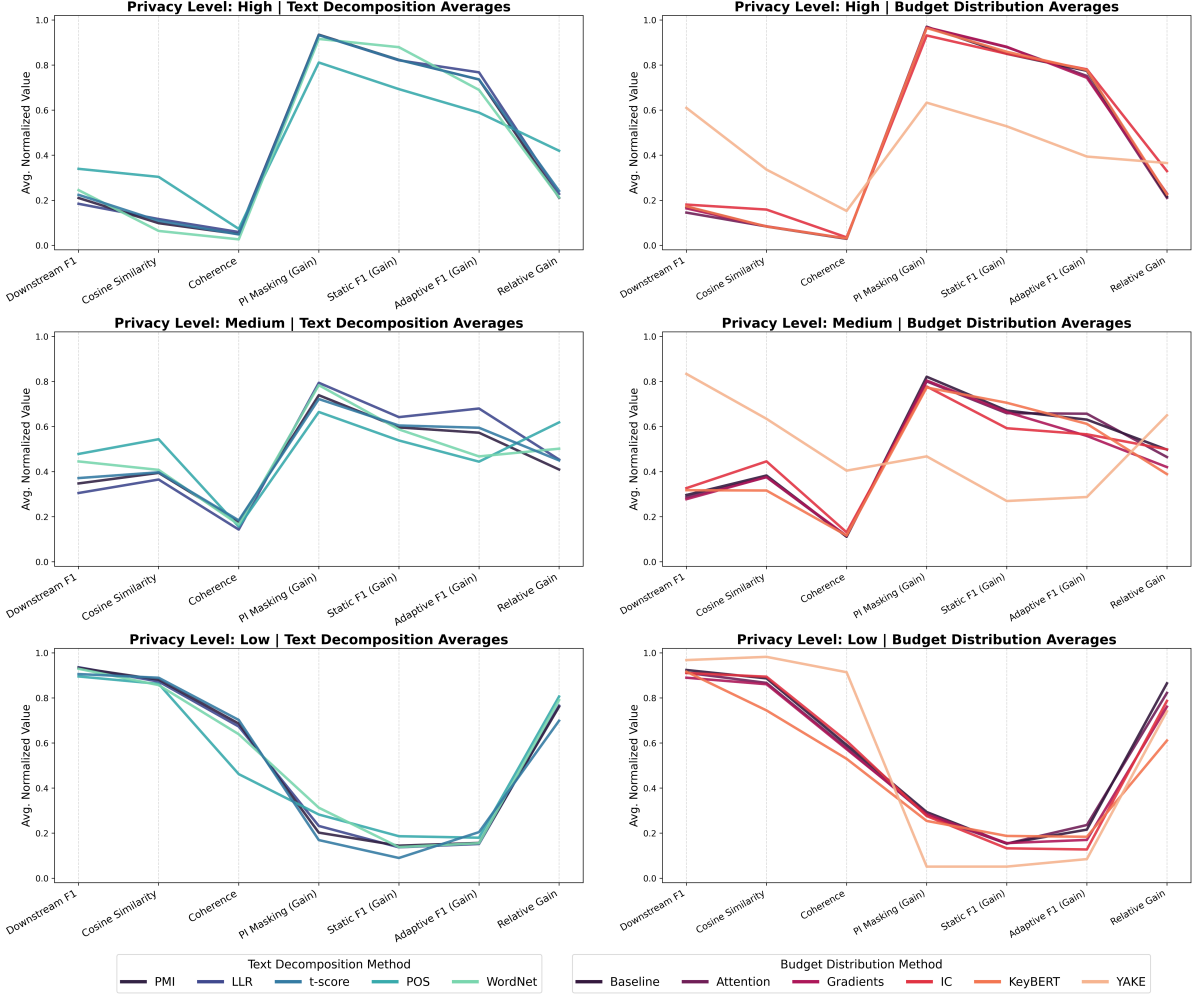


Figure 3: Averaged results over both datasets, for the three selected privacy levels (ϵ budgets). The graphs illustrate the average performance of decomposition and budget distribution over the seven captured metrics. For the privacy metrics, the *gain* (i.e., 1 - score) is presented. All scores are normalized between 0-1, with 1 being the highest score.

et al., 2023). The resulting scores from the three models are averaged for the final similarity score.

Text coherence can be approximated by *perplexity* to give a sense of the quality of obfuscated text output (Weggenmann et al., 2022; Mattern et al., 2022). We report the average perplexity over the first 32 tokens of all texts in a dataset, calculated using a GPT-2 model (Radford et al., 2019).

4.3.3 Trade-off Calculation

To represent the privacy-utility trade-off, we calculate the *relative gain* metric, introduced by Mattern et al. (2022). This metric directly weighs utility losses and privacy gains, and a positive result implies that gained privacy outweighs lost utility. Specifically, we define $RG = \frac{U_p}{U_o} - \frac{P_p}{P_o}$, where U denotes the average utility scores (downstream, similarity), P the average privacy scores (PI, static, adaptive), and the subscripts p and o denote the pri-

vate and original datasets, respectively. Coherence was excluded due to the large/unbounded values.

5 Results and Statistical Analysis

The complete results of all 180 evaluation setups are presented in Tables 6-11 of Appendix D. In Figure 3, we illustrate the aggregated (average) results over both evaluation datasets, for all three privacy levels and for both decomposition and distribution.

Figure 3 clearly shows that despite equal document-level ϵ budgets, different results can be obtained for privacy, utility, and relative gains. Taking the relative gain (trade-off) as a dependent variable, we conduct significance tests to determine if the choice of decomposition or distribution method affects the resulting trade-offs. For this, a two-way ANOVA test (Fisher, 1992) is fitting, as we are studying two categorical independent variables (decomposition and distribution) and one contin-

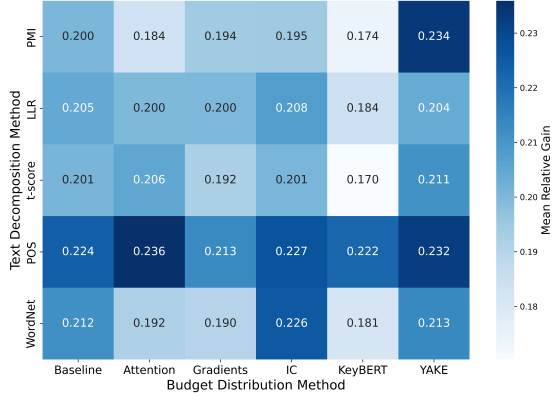


Figure 4: Global relative gain averages ($\uparrow$), i.e., over two datasets and three privacy levels.

uous dependent variable (relative gain). We also conduct one-way tests on the effect of decomposition *or* distribution. All tests are performed using STATSMODEL. The categorical variables *dataset* and *privacy level* are included as controls.

We find that the choice of both decomposition ($F = 5.57, p < 0.001$) and distribution ($F = 3.97, p = 0.002$) has significant effects on the resulting relative gain; however, the *interaction* between the two, as concluded by the two-way test, is not significant ($F = 0.63, p = 0.88$). While the choice of distribution method provides the largest absolute relative gain gap (0.0324 vs. 0.0290) over decomposition, the higher F -statistic implies that decomposition provides a more consistent impact on relative gain. This is confirmed by a partial eta squared (effect size) value of $\eta_p^2 = 0.132$ for decomposition and $\eta_p^2 = 0.119$ for distribution.

To investigate further, we normalize all scores by the mean of each (*dataset, privacy level*) group to remove confounding effects. Then, we perform a Tukey’s Honestly Significant Difference (HSD) post-hoc test (Tukey, 1953), which tests for significance between all decomposition-distribution pairs (435 in total, or $\frac{(30 \times 30)}{2} - \frac{30}{2}$). The HSD analysis reveals significant differences ($p < 0.05$) between **25** pairs. Notably, the combination of POS+attention achieved the largest significant differences (in magnitude), followed by PMI+YAKE and POS+YAKE. Conversely, 21 of these 25 differences feature KeyBERT in the lower end of the comparison. These results are supported by an illustration of the global averages for relative gain, depicted in Figure 4.

6 Discussion

We reflect on the main findings of our systematic evaluation and discuss their implications.

Break it down and distribute. The findings from our experiments and the resulting analysis demonstrate the importance of selecting appropriate methods for text decomposition and privacy budget distribution in the context of DP text obfuscation. We discover that these decisions can lead to significant differences in the downstream trade-offs, despite using exactly the same document-level ε budgets over a dataset. Interestingly, while the choice of decomposition and distribution is individually significant, their *interaction* is not (by far), suggesting that there are no conflicting effects between the two stages of DP text obfuscation.

These insights support the importance of informed DP text obfuscation, where “simple” uniform budget distribution rarely produces superior results, and equally as importantly, where the manner in which text documents are chunked is crucial to maximizing utility and privacy. Thus, we provide evidence of the fact that equivalent theoretical privacy guarantees (via ε) does not necessitate equivalent empirical results, and this must be carefully tuned to optimize privacy-utility trade-offs.

What works well together? Looking into potential optimal combinations of decomposition and distribution techniques, as found in our experiment results, reveals interesting insights. Optimizing for relative gains yields the clear choice of POS+Attention, a pair which achieved an average trade-off of 0.236. The combination of PMI+YAKE is the best-performing strategy for downstream utility preservation. When considering all three utility metrics (F1, CS, and Perplexity), the top-five results (averaged across all setups) *always* feature YAKE as the distribution method.

In scenarios where privacy protection is crucial, though, different winners emerge. Averaging the three privacy metrics we employ, the combination of LLR+KEYBERT achieves the best results across all setups. In the top-five average privacy results, LLR and WordNet both appear twice for decomposition, and likewise for KEYBERT and Attention. Thus, while our statistical analysis shows that choice of decomposition and distribution can be considered separately, our results imply that when optimizing for privacy, utility, and their trade-offs, particular strategies may be more well-suited.

The curious case of trade-offs. Looking purely at resulting trade-offs, though, we learn that this metric is not only significantly influenced by decomposition and distribution, but also by dataset

	sum_sq	df	F	p
C(Decomposition)	0.0214	4.0	5.57	3.32e-04
C(Distribution)	0.0191	5.0	3.97	2.10e-03
C(Dataset)	0.0370	1.0	38.44	5.42e-09
Q(Privacy Level)	0.3173	2.0	165.01	2.67e-38
C(Decomposition):C(Distribution)	0.0122	20.0	0.63	8.83e-01
Residual	0.1414	147.0	—	—

Table 2: ANOVA test results, with the sum of squares, degrees of freedom (df), the F -statistic, and the p -value of F . Significant p -values ($p < 0.05$) are **bolded**.

and privacy level effects (as evidenced in the full analysis results in Table 2). The impact of *privacy level*, or ϵ budget, is also made clear in Figure 3, which paradoxically illustrates average relative gains increasing as the privacy level decreases.

This “curious case” calls to question the possible dominance of utility measurement in privacy-utility trade-off calculations. More importantly, it points to the idea that there presumably exists a (local) maximum in terms of trade-offs; however, this must be meticulously balanced such that the boundary case of very high utility preservation and very low privacy gains does not become the optimal result.

Practical implications. The results of our systematic evaluation carry practical implications. While our experiments are limited in scope and do not represent a comprehensive factorial analysis of decomposition and distribution in DP text obfuscation, it does define a blueprint for doing so at scale. We show that with the support of an experiment setup like ours, practitioners can become informed on which particular setups work optimally, for a given data domain and privacy preference. This becomes important for providing explainability and usability to the scalable ϵ parameter, and it allows for flexibility in defining priority objectives, which we assume in this work to be positive trade-offs.

We highlight the need to view the integration of DP and textual data as a *linguistics*-inspired task. This embodies *divide and conquer*, working with text in tandem with considerations of optimal groupings of semantic meaning, as well as the relative quantification of their “importance” in context. With this *divide and conquer* mindset, we argue that DP text privatization can not only be optimized, but also be more aligned to true privacy protection.

7 Conclusion

We conduct a systematic evaluation of five text decomposition methods and six privacy budget distribution methods for document-level DP text obfuscation. Our experiments demonstrate the im-

portance of the design of privatization procedures, where the precise allocation of the privacy budget has significant implications on downstream privacy and utility. As such, we advance the understanding and optimization of DP text privatization, providing a foundation for future work on (1) designing and evaluating intelligent methods for decomposing texts into coherent chunks, (2) devising methods for quantifying the “importance” of such chunks to utility and privacy, and (3) conducting systematic studies on the factors contributing to trade-offs when operating local DP mechanisms on text.

Limitations

Our work is limited by the selected methods for text decomposition and privacy budget distribution, which we selected to be a representative sample of suitable methods from an informal literature review. Future work could build on this foundation to explore a wide breadth of methods, which would also serve to validate our findings.

Since the design and evaluation of DP mechanisms was not central to our work, we only used the original MADLIB implementation from Feyisetan et al. (2020), adapted to function with our various trained embedding models. As such, we cannot make any conclusions regarding the (statistical) effect of mechanism choice, which could be added in future systematic evaluations for a more complete picture of DP text obfuscation effectiveness.

Limitations to our evaluation procedure also include the sole focus on the English language, as well as the limited domain scope (user review texts). Follow-up studies should expand evaluation to other languages and domains for greater generalizability of our findings.

As with any quantification of privacy, utility, and the privacy-utility trade-off, our evaluations are limited by the chosen metrics, which provide a snapshot of what one may consider privacy or utility. Particularly with privacy evaluation, the measure of personal identifier masking and attribute inference detection does not constitute a holistic test of privacy preservation; we use these metrics as a proxy, following previous works in the field.

Ethical Considerations

We confirm that all utilized software and datasets for our experiments are open-source and acceptable to use under their respective licenses. No data was collected from human subjects.

References

- Stefan Arnold, Dilara Yesilbas, and Sven Weinzierl. 2023a. [Driving context into text-to-text privatization](#). In *Proceedings of the 3rd Workshop on Trustworthy Natural Language Processing (TrustNLP 2023)*, pages 15–25, Toronto, Canada. Association for Computational Linguistics.
- Stefan Arnold, Dilara Yesilbas, and Sven Weinzierl. 2023b. [Guiding text-to-text privatization by syntax](#). In *Proceedings of the 3rd Workshop on Trustworthy Natural Language Processing (TrustNLP 2023)*, pages 151–162, Toronto, Canada. Association for Computational Linguistics.
- Doug Beeferman, Adam Berger, and John Lafferty. 1999. [Statistical models for text segmentation](#). *Machine learning*, 34(1):177–210.
- Vishal Bhalla and Klara Klimcikova. 2019. [Evaluation of automatic collocation extraction methods for language learning](#). In *Proceedings of the Fourteenth Workshop on Innovative Use of NLP for Building Educational Applications*, pages 264–274, Florence, Italy. Association for Computational Linguistics.
- Steven Bird, Ewan Klein, and Edward Loper. 2009. *Natural language processing with Python: analyzing text with the natural language toolkit*. O’Reilly Media, Inc.
- Ricardo Campos, Vítor Mangaravite, Arian Pasquali, Alípio Jorge, Célia Nunes, and Adam Jatowt. 2020. [Yake! keyword extraction from single documents using multiple local features](#). *Information Sciences*, 509:257–289.
- Ricardo Silva Carvalho, Theodore Vasiloudis, Oluwaseyi Feyisetan, and Ke Wang. 2023. [TEM: high utility metric differential privacy on text](#). In *Proceedings of the 2023 SIAM International Conference on Data Mining (SDM)*, pages 883–890. SIAM.
- Konstantinos Chatzikokolakis, Miguel E Andrés, Nicolás Emilio Bordenabe, and Catuscia Palamidessi. 2013. [Broadening the scope of differential privacy using metrics](#). In *Privacy Enhancing Technologies: 13th International Symposium, PETS 2013, Bloomington, IN, USA, July 10-12, 2013. Proceedings 13*, pages 82–102. Springer.
- Kenneth Ward Church and Patrick Hanks. 1990. [Word association norms, mutual information, and lexicography](#). *Computational Linguistics*, 16(1):22–29.
- Mathieu Constant, Gülşen Eryiğit, Johanna Monti, Lonneke van der Plas, Carlos Ramisch, Michael Rosner, and Amalia Todirascu. 2017. [Survey: Multiword expression processing: A Survey](#). *Computational Linguistics*, 43(4):837–892.
- Francesco Luigi De Faveri, Guglielmo Faggioli, and Nicola Ferro. 2025. [Dp-comet: A differential privacy contextual obfuscation mechanism for texts in natural language processing](#). In *Proceedings of the 34th ACM International Conference on Information and Knowledge Management, CIKM ’25*, page 4700–4705, New York, NY, USA. Association for Computing Machinery.
- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. [BERT: Pre-training of deep bidirectional transformers for language understanding](#). In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 4171–4186, Minneapolis, Minnesota. Association for Computational Linguistics.
- Ted Dunning. 1993. [Accurate methods for the statistics of surprise and coincidence](#). *Computational Linguistics*, 19(1):61–74.
- Cynthia Dwork. 2006. [Differential privacy](#). In *International colloquium on automata, languages, and programming*, pages 1–12. Springer.
- Natasha Fernandes, Mark Dras, and Annabelle McIver. 2019. [Generalised differential privacy for text document processing](#). In *Principles of Security and Trust: 8th International Conference, POST 2019, Held as Part of the European Joint Conferences on Theory and Practice of Software, ETAPS 2019, Prague, Czech Republic, April 6–11, 2019, Proceedings 8*, pages 123–148. Springer International Publishing.
- Oluwaseyi Feyisetan, Borja Balle, Thomas Drake, and Tom Diethe. 2020. [Privacy- and utility-preserving textual analysis via calibrated multivariate perturbations](#). In *Proceedings of the 13th International Conference on Web Search and Data Mining, WSDM ’20*, page 178–186, New York, NY, USA. Association for Computing Machinery.
- R. A. Fisher. 1992. *Statistical Methods for Research Workers*, pages 66–70. Springer New York, New York, NY.
- Maarten Grootendorst. 2020. [Keybert: Minimal keyword extraction with bert](#).
- Lulu Gu, Yue Pan, and Pengyuan Liu. 2021. [A comparative study of collocation extraction methods from the perspectives of vocabulary and grammar: A case study in the field of journalism](#). In *Proceedings of the 35th Pacific Asia Conference on Language, Information and Computation*, pages 201–210, Shanghai, China. Association for Computational Linguistics.
- Pengcheng He, Xiaodong Liu, Jianfeng Gao, and Weizhu Chen. 2021. [Deberta: Decoding-enhanced bert with disentangled attention](#). *Preprint*, arXiv:2006.03654.
- Dirk Hovy, Anders Johannsen, and Anders Søgaard. 2015. [User review sites as a resource for large-scale sociolinguistic studies](#). In *Proceedings of the 24th International Conference on World Wide Web, WWW*

- '15, page 452–461, Republic and Canton of Geneva, CHE. International World Wide Web Conferences Steering Committee.
- Lijie Hu, Ivan Habernal, Lei Shen, and Di Wang. 2024. [Differentially private natural language models: Recent advances and future directions](#). In *Findings of the Association for Computational Linguistics: EACL 2024*, pages 478–499, St. Julian's, Malta. Association for Computational Linguistics.
- Timour Igamberdiev, Thomas Arnold, and Ivan Habernal. 2022. [DP-rewrite: Towards reproducibility and transparency in differentially private text rewriting](#). In *Proceedings of the 29th International Conference on Computational Linguistics*, pages 2927–2933, Gyeongju, Republic of Korea. International Committee on Computational Linguistics.
- Shiva Prasad Kasiviswanathan, Homin K. Lee, Kobbi Nissim, Sofya Raskhodnikova, and Adam Smith. 2008. [What can we learn privately?](#) In *2008 49th Annual IEEE Symposium on Foundations of Computer Science*, pages 531–540.
- Kunho Kim, Sivakanth Gopi, Janardhan Kulkarni, and Sergey Yekhanin. 2021. [Differentially private n-gram extraction](#). In *Advances in Neural Information Processing Systems*, volume 34, pages 5102–5111. Curran Associates, Inc.
- Oleksandra Klymenko, Stephen Meisenbacher, and Florian Matthes. 2022. [Differential privacy in natural language processing: The story so far](#). In *Proceedings of the Fourth Workshop on Privacy in Natural Language Processing*, pages 1–11, Seattle, United States. Association for Computational Linguistics.
- Zehan Li, Xin Zhang, Yanzhao Zhang, Dingkun Long, Pengjun Xie, and Meishan Zhang. 2023. [Towards general text embeddings with multi-stage contrastive learning](#). *Preprint*, arXiv:2308.03281.
- Darshini Mahendran, Changqing Luo, and Bridget T. McInnes. 2021. [Review: Privacy-preservation in the context of natural language processing](#). *IEEE Access*, 9:147600–147612.
- Justus Mattern, Benjamin Weggenmann, and Florian Kerschbaum. 2022. [The limits of word level differential privacy](#). In *Findings of the Association for Computational Linguistics: NAACL 2022*, pages 867–881, Seattle, United States. Association for Computational Linguistics.
- Casey Meehan, Khalil Mrini, and Kamalika Chaudhuri. 2022. [Sentence-level privacy for document embeddings](#). In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 3367–3380, Dublin, Ireland. Association for Computational Linguistics.
- Stephen Meisenbacher, Maulik Chevli, and Florian Matthes. 2024. [A collocation-based method for addressing challenges in word-level metric differential privacy](#). In *Proceedings of the Fifth Workshop on Privacy in Natural Language Processing*, pages 39–51, Bangkok, Thailand. Association for Computational Linguistics.
- Stephen Meisenbacher, Chaeun Joy Lee, and Florian Matthes. 2025. [Spend your budget wisely: Towards an intelligent distribution of the privacy budget in differentially private text rewriting](#). In *Proceedings of the Fifteenth ACM Conference on Data and Application Security and Privacy, CODASPY '25*, page 84–95, New York, NY, USA. Association for Computing Machinery.
- Tomas Mikolov, Kai Chen, Greg Corrado, and Jeffrey Dean. 2013. [Efficient estimation of word representations in vector space](#). *Preprint*, arXiv:1301.3781.
- George A. Miller. 1995. [Wordnet: a lexical database for english](#). *Commun. ACM*, 38(11):39–41.
- Irina Pak and Phoey Lee Teh. 2017. [Text segmentation techniques: a critical review](#). *Innovative Computing, Optimization and Its Applications: Modelling and Simulations*, pages 167–181.
- Xudong Pan, Mi Zhang, Shouling Ji, and Min Yang. 2020. [Privacy risks of general-purpose language models](#). In *2020 IEEE Symposium on Security and Privacy (SP)*, pages 1314–1331.
- Pavel Pecina and Pavel Schlesinger. 2006. [Combining association measures for collocation extraction](#). In *Proceedings of the COLING/ACL 2006 Main Conference Poster Sessions*, pages 651–658, Sydney, Australia. Association for Computational Linguistics.
- Guilherme Penedo, Hynek Kydlíček, Loubna Ben Al-lal, Anton Lozhkov, Margaret Mitchell, Colin Raffel, Leandro Von Werra, and Thomas Wolf. 2024. [The fineweb datasets: decanting the web for the finest text data at scale](#). In *Proceedings of the 38th International Conference on Neural Information Processing Systems, NIPS '24*, Red Hook, NY, USA. Curran Associates Inc.
- Alec Radford, Jeff Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever. 2019. [Language models are unsupervised multitask learners](#). *OpenAI*.
- Nils Reimers and Iryna Gurevych. 2019. [Sentence-BERT: Sentence embeddings using Siamese BERT-networks](#). In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 3982–3992, Hong Kong, China. Association for Computational Linguistics.
- Ivan A Sag, Timothy Baldwin, Francis Bond, Ann Copestake, and Dan Flickinger. 2002. [Multiword expressions: A pain in the neck for nlp](#). In *International conference on intelligent text processing and computational linguistics*, pages 1–15. Springer.

Samuel Sousa and Roman Kern. 2023. [How to keep text private? a systematic review of deep learning methods for privacy-preserving natural language processing](#). *Artificial Intelligence Review*, 56(2):1427–1492.

Mukund Sundararajan, Ankur Taly, and Qiqi Yan. 2017. [Axiomatic attribution for deep networks](#). In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pages 3319–3328. PMLR.

Erik F. Tjong Kim Sang and Sabine Buchholz. 2000. [Introduction to the CoNLL-2000 shared task chunking](#). In *Fourth Conference on Computational Natural Language Learning and the Second Learning Language in Logic Workshop*.

John W. Tukey. 1953. [Section of mathematics and engineering: Some selected quick and easy methods of statistical analysis](#). *Transactions of the New York Academy of Sciences*, 16(2 Series II):88–97.

Saiteja Utpala, Sara Hooker, and Pin-Yu Chen. 2023. [Locally differentially private document generation using zero shot prompting](#). In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 8442–8457, Singapore. Association for Computational Linguistics.

Benjamin Weggenmann, Valentin Rublack, Michael Andrejczuk, Justus Mattern, and Florian Kerschbaum. 2022. [Dp-vae: Human-readable text anonymization for online reviews with differentially private variational autoencoders](#). In *Proceedings of the ACM Web Conference 2022, WWW ’22*, page 721–731, New York, NY, USA. Association for Computing Machinery.

Yifan Yao, Jinhao Duan, Kaidi Xu, Yuanfang Cai, Zhibo Sun, and Yue Zhang. 2024. [A survey on large language model \(llm\) security and privacy: The good, the bad, and the ugly](#). *High-Confidence Computing*, 4(2):100211.

Xiang Yue, Minxin Du, Tianhao Wang, Yaliang Li, Huan Sun, and Sherman S. M. Chow. 2021. [Differential privacy for text analytics via natural text sanitization](#). In *Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021*, pages 3853–3866, Online. Association for Computational Linguistics.

A Reproducibility

Hardware. All CPU-based experiments were run on a single Intel Xeon Gold 6148 20-core CPU, and all experiments benefiting from GPU acceleration were run on a Nvidia Tesla V100 16GB GPU.

Random Seeding. For any procedure requiring randomization, we used a random seed of 42. For training tasks where we ran three training runs (and reported the average score), we used the seeds of 42, 43, and 44, sequentially.

Training Parameters. The training parameters used for WORD2VEC models are given in Table 3.

For the privacy and utility experiments which involved the fine-tuning of a DEBERTA-V3-BASE, the training parameters are reported in Table 4.

Parameter	Value
Vector Size	300
Architecture	Skip-gram
Training Workers	5
Window Size	5
Minimum Count	5
Training Epochs	5
Negative Samples	5

Table 3: Word2Vec Training Parameters

Parameter	Value
Model	DEBERTA-V3-BASE
Num. Epochs	1
Learning Rate	2e-5
Warmup Steps	500
Batch Size	8
Loss Function	Focal Loss
Focal Loss γ	2.0
Focal Loss α	0.25
Train/Test Split	90% / 10%
Number of Runs	3
Random Seed	42

Table 4: DEBERTA Training Parameters.

B Association Measure Calculation

With N as the total unigram count and c as a frequency count, the trigram PMI score of (w_1, w_2, w_3) is defined as:

$$\text{PMI}(w_1, w_2, w_3) = \log_2 \frac{c(w_1, w_2, w_3) \cdot N^2}{c(w_1) \cdot c(w_2) \cdot c(w_3)}$$

Similarly, the quad-gram PMI score:

$$\text{PMI}(w_1, w_2, w_3, w_4) = \log_2 \frac{c(w_1, w_2, w_3, w_4) \cdot N^3}{c(w_1) \cdot c(w_2) \cdot c(w_3) \cdot c(w_4)}$$

For LLR, the calculated scores are based on a 2x2 contingency table (Table 5), exemplified by:

$$\text{LLR} = 2 \left(x \log x(N) - x \log x(c_{11} + c_{12}) - x \log x(c_{21} + c_{22}) - x \log x(c_{11} + c_{21}) - x \log x(c_{12} + c_{22}) + x \log x(c_{11}) + x \log x(c_{12}) + x \log x(c_{21}) + x \log x(c_{22}) \right)$$

	w_1	Not w_1
w_2	$c(w_1, w_2)$	$c(w_2 \text{ not } w_1)$
Not w_2	$c(w_1 \text{ not } w_2)$	$c(\text{not } w_1 \text{ or } w_2)$

Table 5: Example Contingency Table for LLR Score.

Finally, we define the trigram t-score as:

$$t(w_1, w_2, w_3) = \frac{c(w_1, w_2, w_3) - \frac{c(w_1) \cdot c(w_2) \cdot c(w_3)}{N^2}}{\sqrt{c(w_1, w_2, w_3)}}$$

And the quad-gram t-score as:

$$t(w_1, w_2, w_3, w_4) = \frac{c(w_1, w_2, w_3, w_4) - \frac{c(w_1) \cdot c(w_2) \cdot c(w_3) \cdot c(w_4)}{N^3}}{\sqrt{c(w_1, w_2, w_3, w_4)}}$$

C Algorithm Pseudocode

The algorithms for text decomposition are outlined in Algorithm 1 for association-based methods, Algorithm 2 for POS-based decomposition, and Algorithm 3 for WordNet-based decomposition.

Detailed pseudocode for budget distribution methods can be found for Attention Weights (Algorithm 4), Integrated Gradients (Algorithm 5), and Information Content (Algorithm 6).

D Complete Results

The complete experiment results from Trustpilot with privacy levels of high, medium, and low are presented in Tables 6, 7, and 8, respectively. Likewise, the complete results from Yelp are presented in Tables 9, 10, and 11. As Figure 3 only presents the average scores for both decomposition and distribution, we visualize the averaged results for all 30 combinations in Figure 5.

E Example Texts.

Table 12 shows corresponding text examples from a single original text from the Trustpilot dataset, privatized with all combinations of decomposition and distribution methods, at $\varepsilon = 52$.

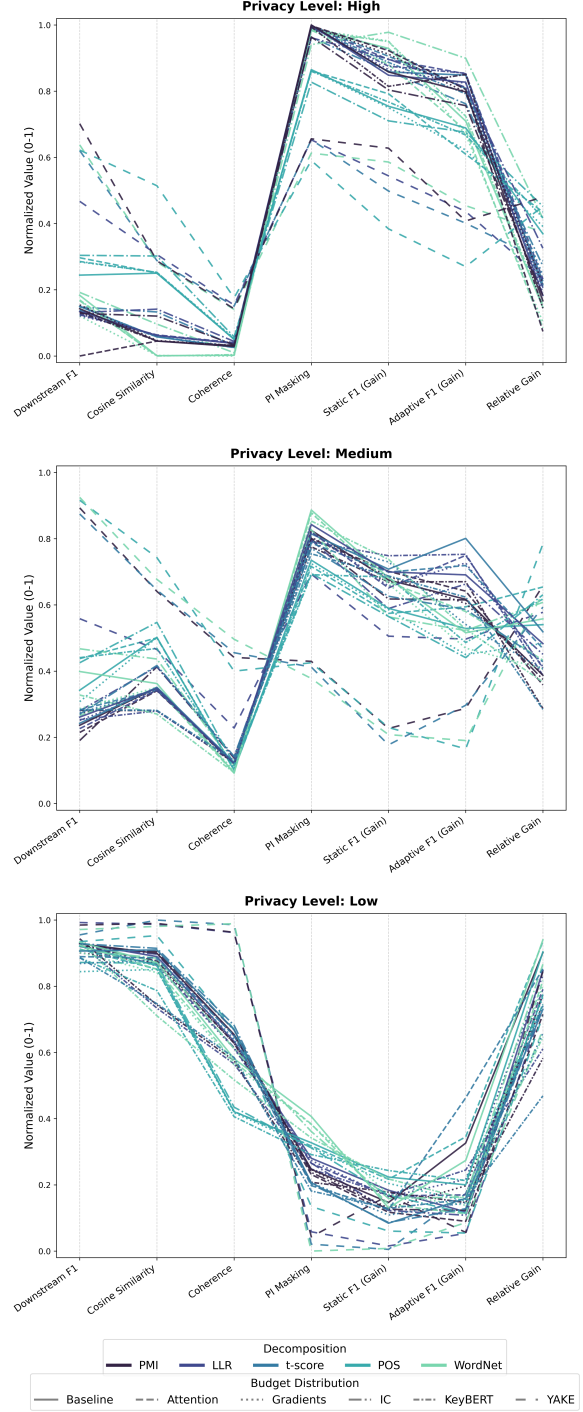


Figure 5: Averaged experiment results over the two selected datasets. Each line represents the mean results for a decomposition-distribution pair, and the y axes plot the normalized scores for all captured metrics. For all privacy metrics, the *gain* (1 - score) is shown, such that for all metrics, 1 represents the best score.

Algorithm 1: Association-based Decomposition

(PMI/LLR/t-score)

Input: document D , n-gram sets $\mathcal{B}$, $\mathcal{T}$, $\mathcal{Q}$, stopwords $\mathcal{S}$
Output: chunk sequence C

```
1  $C \leftarrow []$ ;
2 sentences  $\leftarrow$  SentenceTokenize( $D$ );
3 foreach sentence  $\in$  sentences do
4   tokens  $\leftarrow$  WordTokenize(sentence, pattern =
      \b\w+\b);
5   tokens  $\leftarrow$  Lowercase(tokens);
6    $i \leftarrow 0$ ;
7   while  $i < |tokens|$  do
8     matched  $\leftarrow$  False;
9     if  $i + 3 < |tokens|$  then
10      ngram  $\leftarrow$  Join(tokens[ $i : i + 4$ ], " ");
11      if ngram  $\in \mathcal{Q}$  then
12         $C$ .extend(ProcessStopwords(tokens[ $i : i + 4$ ],  $\mathcal{S}$ ));
13         $i \leftarrow i + 4$ ;
14        matched  $\leftarrow$  True;
15        continue;
16      if  $\neg$ matched  $\wedge i + 2 < |tokens|$  then
17        ngram  $\leftarrow$  Join(tokens[ $i : i + 3$ ], " ");
18        if ngram  $\in \mathcal{T}$  then
19           $C$ .extend(ProcessStopwords(tokens[ $i : i + 3$ ],  $\mathcal{S}$ ));
20           $i \leftarrow i + 3$ ;
21          matched  $\leftarrow$  True;
22          continue;
23      if  $\neg$ matched  $\wedge i + 1 < |tokens|$  then
24        ngram  $\leftarrow$  Join(tokens[ $i : i + 2$ ], " ");
25        if ngram  $\in \mathcal{B}$  then
26           $C$ .extend(ProcessStopwords(tokens[ $i : i + 2$ ],  $\mathcal{S}$ ));
27           $i \leftarrow i + 2$ ;
28          matched  $\leftarrow$  True;
29          continue;
30       $C$ .append(tokens[ $i$ ]);
31       $i \leftarrow i + 1$ ;
32  $C \leftarrow$  MergeContractions( $C$ );
33 return  $C$ ;
```

Algorithm 2: POS-based Decomposition with BigramTagger

Input: document D , trained BigramTagger θ , stopwords $\mathcal{S}$
Output: chunk sequence C

```
1  $C \leftarrow []$ ;
2 sentences  $\leftarrow$  SentenceTokenize( $D$ );
3 foreach sentence  $\in$  sentences do
4   tokens  $\leftarrow$  WordTokenize(sentence);
5   tokens  $\leftarrow$  Lowercase(tokens);
6   pos_tags  $\leftarrow$  POSTagger(tokens);
7   chunk_tags  $\leftarrow \theta$ .tag(pos_tags);
8   tree  $\leftarrow$  ParseIOB(tokens, pos_tags, chunk_tags);
9   foreach subtree  $\in$  tree do
10    if IsChunk(subtree) then
11      words  $\leftarrow$  ExtractWords(subtree);
12      processed  $\leftarrow$  ProcessStopwords(words,  $\mathcal{S}$ );
13       $C$ .extend(processed);
14    else
15      word  $\leftarrow$  ExtractWord(subtree);
16       $C$ .append(word);
17  $C \leftarrow$  MergeContractions( $C$ );
18 return  $C$ ;
```

Algorithm 3: WordNet-based Decomposition

Input: document D , WordNet database $\mathcal{W}$, stopwords $\mathcal{S}$
Output: chunk sequence C

```
1  $C \leftarrow []$ ;
2 sentences  $\leftarrow$  SentenceTokenize( $D$ );
3 foreach sentence  $\in$  sentences do
4   tokens  $\leftarrow$  WordTokenize(sentence);
5   tokens  $\leftarrow$  Lowercase(tokens);
6    $i \leftarrow 0$ ;
7   while  $i < |tokens|$  do
8     matched  $\leftarrow$  False;
9     if  $i + 3 < |tokens|$  then
10      lemma  $\leftarrow$  Join(tokens[ $i : i + 4$ ], "_");
11      if  $\mathcal{W}.$ synsets(lemma)  $\neq \emptyset$  then
12         $C$ .extend(ProcessStopwords(tokens[ $i : i + 4$ ],  $\mathcal{S}$ ));
13         $i \leftarrow i + 4$ ;
14        matched  $\leftarrow$  True;
15        continue;
16      if  $\neg$ matched  $\wedge i + 2 < |tokens|$  then
17        lemma  $\leftarrow$  Join(tokens[ $i : i + 3$ ], "_");
18        if  $\mathcal{W}.$ synsets(lemma)  $\neq \emptyset$  then
19           $C$ .extend(ProcessStopwords(tokens[ $i : i + 3$ ],  $\mathcal{S}$ ));
20           $i \leftarrow i + 3$ ;
21          matched  $\leftarrow$  True;
22          continue;
23      if  $\neg$ matched  $\wedge i + 1 < |tokens|$  then
24        lemma  $\leftarrow$  Join(tokens[ $i : i + 2$ ], "_");
25        if  $\mathcal{W}.$ synsets(lemma)  $\neq \emptyset$  then
26           $C$ .extend(ProcessStopwords(tokens[ $i : i + 2$ ],  $\mathcal{S}$ ));
27           $i \leftarrow i + 2$ ;
28          matched  $\leftarrow$  True;
29          continue;
30       $C$ .append(tokens[ $i$ ]);
31       $i \leftarrow i + 1$ ;
32  $C \leftarrow$  MergeContractions( $C$ );
33 return  $C$ ;
```

Algorithm 4: Extract Attention Weights

Input: text T , privacy budget ϵ
Output: token-budget pairs (w_i, ϵ_i)

```
1 tokens  $\leftarrow$  BertTokenizer( $T$ );
2 inputs  $\leftarrow$  prepare_inputs(tokens);
3 outputs  $\leftarrow$  BertModel(inputs);
4  $\mathcal{A} \leftarrow$  outputs.attentions;
5  $\mathcal{A}_{\text{avg}} \leftarrow$  mean( $\mathcal{A}$ , dim = (layers, heads));
6 scores  $\leftarrow$  mean( $\mathcal{A}_{\text{avg}}$ , dim = tokens);
7 scored_tokens  $\leftarrow$  zip(tokens, scores);
8 combined  $\leftarrow$  combine_subwords(scored_tokens);
9 filtered  $\leftarrow$  remove(combined, special_tokens);
10 budgets  $\leftarrow$  normalize_and_distribute(filtered,  $\epsilon$ );
11 return budgets;
```

Algorithm 5: Extract Integrated Gradients

Input: text T , privacy budget ϵ
Output: token-budget pairs (w_i, ϵ_i)

```
1 tokens  $\leftarrow$  BertTokenizer( $T$ );
2  $E \leftarrow$  BertEmbeddings(tokens);
3 Function ForwardFunc( $E$ ):
4    $H \leftarrow$  BertModel(inputs_embeds =  $E$ );
5   return mean( $H$ ).sum();
6 attributions  $\leftarrow$  IntegratedGradients(ForwardFunc,  $E$ );
7 scores  $\leftarrow$  ||attributions||2;
8 scored_tokens  $\leftarrow$  zip(tokens, scores);
9 combined  $\leftarrow$  combine_subwords(scored_tokens);
10 filtered  $\leftarrow$  remove(combined, special_tokens);
11 budgets  $\leftarrow$  normalize_and_distribute(filtered,  $\epsilon$ );
12 return budgets;
```

Algorithm 6: Get Information Content

Input: text T , privacy budget ϵ
Output: token-budget pairs (w_i, ϵ_i)

```
1 tokens  $\leftarrow$  tokenize( $T$ );
2 pos_tags  $\leftarrow$  pos_tag(tokens);
3 foreach  $(w_i, pos_i) \in pos\_tags$  do
4   wn_pos  $\leftarrow$  convert_to_wordnet(pos $i$ );
5   sense  $\leftarrow$  Lesk(tokens,  $w_i$ , wn_pos);
6   if sense  $\neq$  null then
7     ic_values  $\leftarrow$  [];
8     foreach corpus  $\in IC\_corpora$  do
9       ic_values.append(IC(sense, corpus));
10    score $i$   $\leftarrow$  mean(ic_values);
11  else
12    score $i$   $\leftarrow$  1.0;
13 filtered  $\leftarrow$  filter_alphanumeric(scored_tokens);
14 budgets  $\leftarrow$  normalize_and_distribute(filtered,  $\epsilon$ );
15 return budgets;
```

Algorithm 7: Convert Scores to Budget Distribution

Input: scored tokens (t_i, s_i) , original text T , privacy budget ϵ , invert flag
Output: token-budget pairs (w_i, ϵ_i)

```
1 original_words  $\leftarrow$  tokenize( $T$ );
2 score_map  $\leftarrow$  dict(scored_tokens);
3 scores  $\leftarrow$  [];
4 foreach word  $\in original\_words$  do
5   if word  $\in stopwords$  then
6     scores.append(0.0);
7   else
8     score  $\leftarrow$  score_map.get(word, 0.0);
9     scores.append(score);
10 if  $\exists s \in scores : s < 0$  then
11   min_val  $\leftarrow$  min( $\{s \mid s \neq 0\}$ );
12   foreach  $s_i \in scores$  where  $s_i \neq 0$  do
13      $s_i \leftarrow s_i + |min\_val|$ ;
14 if invert = True then
15   non_zero_scores  $\leftarrow$   $\{s \mid s > 0\}$ ;
16   min_nz  $\leftarrow$  min(non_zero_scores);
17   max_nz  $\leftarrow$  max(non_zero_scores);
18   if max_nz = min_nz then
19     inverted  $\leftarrow$  ones(length(non_zero_scores));
20   else
21     inverted  $\leftarrow$  (max_nz + min_nz) - non_zero_scores;
22   budgets  $\leftarrow$  (inverted /  $\sum$  inverted)  $\times \epsilon$ ;
23 else
24   total  $\leftarrow \sum scores$ ;
25   if total > 0 then
26     budgets  $\leftarrow$  (scores/total)  $\times \epsilon$ ;
27 return zip(original_words, budgets);
```

Algorithm 8: Get Chunked Budgets

Input: clean text T_{clean} , chunker, distributor, privacy budget ϵ
Output: chunk-budget pairs (c_i, ϵ_i) , chunks list C

```
1 chunked_sentences  $\leftarrow$  chunker.chunk( $T_{clean}$ );
2  $C' \leftarrow$  flatten(chunked_sentences);
3 word_budgets  $\leftarrow$  distributor.distribute( $T_{clean}$ ,  $\epsilon$ );
4 budget_iter  $\leftarrow$  iterator(word_budgets);
5 chunk_budgets  $\leftarrow$  [];
6 foreach chunk  $\in C$  do
7   words  $\leftarrow$  split(chunk, ' ');
8    $n \leftarrow$  |words|;
9    $\epsilon_{chunk} \leftarrow$  0.0;
10  for  $i \leftarrow 1$  to  $n$  do
11     $(w, \epsilon_w) \leftarrow$  next(budget_iter);
12     $\epsilon_{chunk} \leftarrow \epsilon_{chunk} + \epsilon_w$ ;
13  chunk_budgets.append((chunk,  $\epsilon_{chunk}$ ));
14 return chunk_budgets,  $C'$ ;
```

Trustpilot $\varepsilon = 5.2$		Budget Distribution						Average
Decomposition	Baseline	Baseline	Attention	Gradients	IC	KeyBERT	YAKE	
		Utility: 98.2 _{0.4} / Adversary: 72.1 _{1.0}						
PMI	Utility F1	76.1 _{2.3}	67.2 _{16.7}	76.8 _{2.8}	75.2 _{2.1}	75.5 _{5.5}	88.6 _{0.9}	76.6
	CS	0.364	0.363	0.365	0.421	0.364	0.425	0.384
	Perplexity	1497	1506	1499	1414	1492	1099	1418
	PI Masking	0.76	0.80	0.78	2.88	0.95	8.27	2.41
	Static F1	43.6 _{4.3}	39.0 _{0.5}	40.0 _{1.4}	47.1 _{5.5}	47.0 _{5.9}	41.1 _{3.5}	43.0
	Adaptive F1	55.5 _{3.8}	56.6 _{1.7}	56.3 _{1.2}	57.2 _{2.3}	53.7 _{2.0}	57.9 _{1.2}	56.2
	Relative Gain	0.159	0.127	0.175	0.153	0.149	0.223₂	0.164
LLR	Utility F1	75.6 _{0.3}	75.0 _{0.5}	74.8 _{3.6}	75.4 _{0.9}	75.1 _{1.4}	88.4 _{0.8}	77.4
	CS	0.383	0.384	0.385	0.442	0.384	0.443	0.404
	Perplexity	1348	1356	1345	1273	1325	1010	1276
	PI Masking	0.79	0.84	0.93	2.96	1.03	8.31	2.48
	Static F1	45.1 _{4.1}	42.1 _{3.9}	44.0 _{5.2}	41.6 _{2.8}	43.0 _{4.9}	47.5 _{6.0}	43.9
	Adaptive F1	56.7 _{2.2}	55.1 _{0.8}	55.9 _{1.9}	55.5 _{2.9}	55.7 _{1.3}	58.3 _{0.8}	56.2
	Relative Gain	0.155	0.171	0.159	0.194	0.165	0.203₂	0.174
t-score	Utility F1	76.5 _{1.7}	75.8 _{5.0}	75.9 _{1.8}	76.4 _{3.6}	76.6 _{1.8}	89.5 _{0.7}	78.5
	CS	0.366	0.366	0.366	0.421	0.366	0.425	0.385
	Perplexity	1479	1487	1467	1404	1474	1061	1395
	PI Masking	0.85	0.88	0.86	2.90	1.05	8.27	2.47
	Static F1	44.7 _{4.2}	41.3 _{2.1}	40.2 _{1.0}	44.0 _{1.8}	42.5 _{2.7}	49.6 _{3.4}	43.7
	Adaptive F1	54.6 _{0.6}	55.7 _{1.2}	55.5 _{2.9}	57.0 _{0.8}	57.0 _{2.4}	59.0 _{1.9}	56.5
	Relative Gain	0.161	0.166	0.172	0.173	0.159	0.187₂	0.170
POS	Utility F1	82.5 _{2.2}	85.7 _{1.0}	84.9 _{1.7}	86.2 _{0.9}	85.0 _{1.4}	92.6 _{0.4}	86.2
	CS	0.504	0.504	0.505	0.545	0.506	0.574	0.523
	Perplexity	1356	1368	1373	1267	1362	992	1286
	PI Masking	5.81	5.87	5.87	8.13	6.01	11.36	7.17
	Static F1	49.0 _{1.9}	46.7 _{3.6}	49.7 _{3.5}	51.5 _{3.1}	48.1 _{9.1}	55.6 _{7.5}	50.1
	Adaptive F1	58.8 _{1.1}	58.4 _{0.7}	58.9 _{3.0}	57.0 _{3.8}	58.5 _{1.9}	63.1 _{1.2}	59.1
	Relative Gain	0.205₁	0.233_{1,2}	0.215₁	0.232	0.223₁	0.224	0.222
WordNet	Utility F1	78.7 _{1.4}	77.6 _{2.6}	74.8 _{1.9}	79.2 _{4.3}	77.8 _{1.7}	90.2 _{1.2}	79.7
	CS	0.336	0.335	0.336	0.413	0.335	0.418	0.362
	Perplexity	2063	2065	2078	1850	2034	1375	1911
	PI Masking	1.44	1.49	1.28	4.25	1.48	9.57	3.25
	Static F1	38.7 _{2.8}	42.1 _{7.5}	37.5 _{0.4}	35.3 _{5.1}	37.6 _{0.3}	40.8 _{1.8}	38.7
	Adaptive F1	57.6 _{0.3}	57.1 _{1.2}	57.0 _{1.4}	49.8 _{12.8}	57.2 _{0.1}	54.0 _{8.4}	55.4
	Relative Gain	0.166	0.148	0.154	0.242_{1,2}	0.167	0.239₂	0.186
Average	Utility F1	77.9	76.3	77.4	78.5	78.0	89.9	–
	CS	0.391	0.390	0.391	0.448	0.391	0.457	–
	Perplexity	1549	1557	1552	1441	1537	1107	–
	PI Masking	1.93	1.98	1.94	4.22	2.10	9.15	–
	Static F1	44.2	42.3	42.3	43.9	43.6	46.9	–
	Adaptive F1	56.7	56.6	56.7	55.3	56.4	58.5	–
	Relative Gain	0.169	0.169	0.175	0.199	0.173	0.215	–

Table 6: Results for the Trustpilot dataset at $\varepsilon = 5.2$. Each combination of decomposition and distribution strategy yields scores for Utility F1 ($\uparrow$), Cosine Similarity (CS, $\uparrow$), Perplexity ($\downarrow$), PI Masking ($\downarrow$), Static F1 ($\downarrow$), Adaptive F1 ($\downarrow$), and Relative Gain ($\uparrow$). Values derived from an average of three runs are displayed as mean_{std}. F1 values are in %. The rightmost column and bottom rows calculate the average results over decomposition and distribution methods, respectively. Best average Relative Gain values are **bolded**. For individual Relative Gain values, only the best are bolded: x_1 for best per decomposition, x_2 for best per budget distribution, and $x_{1,2}$ if both. The global best Relative Gain (across all combinations) is underlined.

Trustpilot $\varepsilon = 52$		Budget Distribution						Average
Decomposition	Baseline	Baseline	Attention	Gradients	IC	KeyBERT	YAKE	
		Utility: 98.2 _{0.4} / Adversary: 72.1 _{1.0}						
PMI	Utility F1	82.0 _{0.8}	80.6 _{0.6}	84.0 _{1.6}	79.1 _{0.5}	82.4 _{2.6}	93.8 _{1.2}	83.6
	CS	0.525	0.521	0.521	0.577	0.492	0.614	0.542
	Perplexity	801	794	787	706	770	515	729
	PI Masking	6.19	7.45	7.40	7.80	9.02	14.11	8.66
	Static F1	49.2 _{2.0}	48.0 _{1.8}	48.8 _{2.0}	52.5 _{0.8}	49.5 _{1.9}	56.3 _{2.9}	50.7
	Adaptive F1	58.9 _{0.8}	59.6 _{0.9}	59.6 _{2.2}	58.9 _{2.9}	58.6 _{0.8}	60.5 _{4.7}	59.4
	Relative Gain	0.211	0.199	0.213	0.202	0.184	0.247₂	0.209
LLR	Utility F1	83.5 _{2.0}	81.3 _{1.6}	82.9 _{1.8}	81.7 _{3.9}	83.0 _{3.6}	94.4 _{0.3}	84.5
	CS	0.529	0.527	0.527	0.583	0.496	0.616	0.546
	Perplexity	765	763	753	681	746	495	701
	PI Masking	6.00	6.81	6.92	8.49	8.46	13.74	8.40
	Static F1	48.2 _{1.3}	51.3 _{5.1}	50.5 _{1.4}	54.9 _{0.7}	44.4 _{6.4}	56.4 _{4.0}	51.0
	Adaptive F1	55.3 _{5.0}	53.6 _{8.1}	56.0 _{3.4}	58.2 _{3.4}	54.3 _{2.8}	60.5 _{4.8}	56.3
	Relative Gain	0.240	0.219	0.220	0.209	0.231	0.252₂	0.228
t-score	Utility F1	82.3 _{2.1}	84.9 _{2.2}	84.3 _{1.7}	84.1 _{0.5}	82.6 _{0.6}	94.3 _{1.3}	85.4
	CS	0.525	0.522	0.522	0.580	0.492	0.616	0.543
	Perplexity	763	758	754	683	730	493	697
	PI Masking	6.66	7.95	8.05	9.36	9.92	14.89	9.47
	Static F1	47.3 _{0.8}	48.0 _{0.6}	46.2 _{8.3}	52.0 _{1.9}	48.0 _{2.6}	59.3 _{1.3}	50.1
	Adaptive F1	50.1 _{12.7}	54.7 _{6.0}	60.4 _{0.4}	60.2 _{1.6}	57.6 _{1.3}	60.7 _{3.5}	57.3
	Relative Gain	0.254_{1,2}	0.239	0.220	0.219	0.192	0.234	0.226
POS	Utility F1	88.6 _{1.7}	90.6 _{1.2}	86.4 _{0.4}	88.1 _{0.5}	85.6 _{1.8}	95.7 _{1.0}	89.2
	CS	0.618	0.618	0.618	0.655	0.603	0.702	0.636
	Perplexity	898	882	869	792	841	534	803
	PI Masking	10.42	10.98	10.83	12.47	12.40	16.13	12.20
	Static F1	53.6 _{2.6}	54.3 _{2.7}	54.0 _{2.8}	54.8 _{2.9}	47.0 _{8.8}	57.7 _{2.3}	53.6
	Adaptive F1	63.7 _{0.3}	58.1 _{3.5}	62.5 _{0.7}	61.6 _{0.5}	62.6 _{0.7}	65.8 _{0.7}	62.4
	Relative Gain	0.236	0.264₁	0.227	0.248	0.237₁	0.265_{1,2}	0.246
WordNet	Utility F1	85.7 _{1.3}	83.8 _{1.7}	85.1 _{1.5}	88.8 _{1.3}	83.4 _{0.9}	94.8 _{1.1}	86.9
	CS	0.529	0.522	0.523	0.595	0.482	0.642	0.549
	Perplexity	939	942	932	793	931	476	836
	PI Masking	4.09	4.32	4.29	7.87	5.67	17.46	7.29
	Static F1	46.7 _{1.3}	48.3 _{2.4}	48.0 _{1.9}	53.9 _{4.9}	43.1 _{1.6}	56.0 _{1.3}	49.3
	Adaptive F1	61.0 _{0.7}	58.0 _{2.2}	60.3 _{1.3}	59.4 _{1.9}	60.8 _{1.0}	63.6 _{3.6}	60.5
	Relative Gain	0.242	0.233	0.232₁	0.252_{1,2}	0.216	0.241	0.236
Average	Utility F1	84.4	84.2	84.5	84.4	83.4	94.6	—
	CS	0.545	0.542	0.542	0.598	0.513	0.638	—
	Perplexity	833	828	819	731	803	503	—
	PI Masking	6.67	7.50	7.50	9.20	9.09	15.27	—
	Static F1	49.0	50.0	49.5	53.6	46.4	57.2	—
	Adaptive F1	57.8	56.8	59.8	59.7	58.8	62.2	—
	Relative Gain	0.236	0.231	0.222	0.226	0.212	0.248	—

Table 7: Results for the Trustpilot dataset at $\varepsilon = 52$. Each combination of decomposition and distribution strategy yields scores for Utility F1 ($\uparrow$), Cosine Similarity (CS, $\uparrow$), Perplexity ($\downarrow$), PI Masking ($\downarrow$), Static F1 ($\downarrow$), Adaptive F1 ($\downarrow$), and Relative Gain ($\uparrow$). Values derived from an average of three runs are displayed as mean_{std}. F1 values are in %. The rightmost column and bottom rows calculate the average results over decomposition and distribution methods, respectively. Best average Relative Gain values are **bolded**. For individual Relative Gain values, only the best are bolded: x_1 for best per decomposition, x_2 for best per budget distribution, and $x_{1,2}$ if both. The global best Relative Gain (across all combinations) is underlined.

Trustpilot $\varepsilon = 260$		Budget Distribution						Average
Decomposition	Baseline	Baseline	Attention	Gradients	IC	KeyBERT	YAKE	
		Utility: 98.2 _{0.4} / Adversary: 72.1 _{1.0}						
PMI	Utility F1	96.3 _{0.4}	96.5 _{0.4}	96.1 _{0.7}	95.7 _{1.3}	97.1 _{0.0}	98.4 _{0.4}	96.7
	CS	0.855	0.841	0.835	0.864	0.759	0.891	0.841
	Perplexity	220	230	230	212	245	171	218
	PI Masking	29.72	30.62	30.14	30.07	31.65	35.48	31.28
	Static F1	64.7 _{3.7}	67.8 _{0.3}	66.4 _{1.9}	66.0 _{0.6}	63.8 _{3.0}	57.5 _{17.8}	64.4
	Adaptive F1	61.4 _{9.3}	69.3 _{1.0}	67.1 _{3.9}	69.5 _{0.9}	66.8 _{1.4}	69.1 _{1.4}	67.2
	Relative Gain	0.280	0.225	0.236	0.241	0.208	0.282_{1,2}	0.245
LLR	Utility F1	95.5 _{0.3}	96.1 _{0.9}	95.6 _{0.3}	96.1 _{0.3}	96.7 _{0.5}	98.0 _{0.5}	96.3
	CS	0.849	0.835	0.828	0.860	0.755	0.888	0.836
	Perplexity	226	232	235	217	246	171	221
	PI Masking	28.23	29.22	28.91	28.94	30.07	34.30	29.94
	Static F1	62.6 _{8.4}	65.2 _{2.1}	64.7 _{1.0}	66.5 _{1.3}	64.9 _{1.5}	68.0 _{0.6}	65.3
	Adaptive F1	68.2 _{1.9}	67.4 _{1.9}	64.0 _{2.5}	70.1 _{1.1}	65.4 _{2.7}	68.5 _{4.1}	67.3
	Relative Gain	0.260₂	0.243	0.255₁	0.241	0.212	0.243	0.242
t-score	Utility F1	96.3 _{1.1}	95.7 _{1.1}	94.9 _{0.4}	97.5 _{0.3}	96.6 _{0.9}	98.4 _{0.6}	96.6
	CS	0.863	0.848	0.840	0.872	0.763	0.899	0.848
	Perplexity	215	220	226	207	241	166	212
	PI Masking	31.90	32.28	32.13	32.05	33.36	36.87	33.10
	Static F1	68.3 _{1.3}	67.0 _{0.6}	67.6 _{1.2}	68.1 _{1.4}	66.9 _{0.7}	68.5 _{1.3}	67.7
	Adaptive F1	67.4 _{1.8}	56.9 _{18.8}	68.1 _{0.8}	69.1 _{1.7}	67.4 _{2.3}	64.0 _{9.1}	65.5
	Relative Gain	0.235	0.271_{1,2}	0.216	0.239	0.186	0.257	0.234
POS	Utility F1	97.3 _{0.7}	96.5 _{0.3}	94.7 _{0.4}	95.9 _{0.9}	96.6 _{0.5}	97.0 _{1.0}	96.3
	CS	0.829	0.822	0.821	0.837	0.784	0.868	0.827
	Perplexity	352	349	354	336	356	269	336
	PI Masking	26.84	27.20	27.70	28.11	29.09	32.14	28.51
	Static F1	64.1 _{1.5}	65.0 _{2.1}	66.1 _{0.8}	67.5 _{0.4}	63.3 _{2.6}	66.4 _{2.5}	65.4
	Adaptive F1	67.5 _{1.6}	62.5 _{8.5}	68.5 _{1.2}	69.0 _{0.1}	66.5 _{2.5}	70.0 _{0.1}	67.3
	Relative Gain	0.261	0.268₂	0.228	0.232	0.233₁	0.237	0.243
WordNet	Utility F1	96.7 _{0.4}	97.5 _{0.4}	95.9 _{0.7}	96.1 _{0.8}	96.6 _{0.2}	97.5 _{0.4}	96.7
	CS	0.833	0.820	0.810	0.849	0.735	0.881	0.821
	Perplexity	257	260	267	238	281	169	245
	PI Masking	22.26	23.54	24.09	25.20	25.96	37.93	26.50
	Static F1	65.7 _{0.7}	66.1 _{1.8}	63.5 _{2.3}	63.8 _{2.6}	61.3 _{1.9}	68.0 _{0.8}	64.7
	Adaptive F1	63.2 _{10.2}	69.0 _{1.7}	69.1 _{2.0}	68.1 _{1.5}	67.7 _{1.4}	66.6 _{4.1}	67.3
	Relative Gain	0.289_{1,2}	0.256	0.251	0.270₁	0.224	0.230	0.253
Average	Utility F1	96.4	96.5	95.4	96.2	96.7	97.9	–
	CS	0.846	0.833	0.827	0.856	0.759	0.885	–
	Perplexity	254	258	262	242	274	189	–
	PI Masking	27.79	28.57	28.59	28.87	30.02	35.34	–
	Static F1	65.1	66.2	65.7	66.4	64.1	65.7	–
	Adaptive F1	65.5	65.0	67.4	69.2	66.8	67.7	–
	Relative Gain	0.265	0.253	0.237	0.245	0.213	0.250	–

Table 8: Results for the Trustpilot dataset at $\varepsilon = 260$. Each combination of decomposition and distribution strategy yields scores for Utility F1 ($\uparrow$), Cosine Similarity (CS, $\uparrow$), Perplexity ($\downarrow$), PI Masking ($\downarrow$), Static F1 ($\downarrow$), Adaptive F1 ($\downarrow$), and Relative Gain ($\uparrow$). Values derived from an average of three runs are displayed as mean_{std} . F1 values are in %. The rightmost columns and bottom rows calculate the average results over decomposition and distribution methods, respectively. Best average Relative Gain values are **bolded**. For individual Relative Gain values, only the best are bolded: x_1 for best per decomposition, x_2 for best per budget distribution, and $x_{1,2}$ if both. The global best Relative Gain (across all combinations) is underlined.

Yelp $\varepsilon = 18.7$		Budget Distribution						Average
Decomposition	Baseline	Baseline	Attention	Gradients	IC	KeyBERT	YAKE	
		Utility: 87.8 _{1.9} / Adversary: 94.4 _{0.2}						
PMI	Utility F1	48.1 _{0.0}	48.1 _{0.0}	48.1 _{0.0}	48.1 _{0.0}	48.1 _{0.0}	76.6 _{3.2}	52.8
	CS	0.348	0.349	0.349	0.380	0.348	0.590	0.394
	Perplexity	717	692	693	702	687	322	635
	PI Masking	0.37	0.48	0.57	0.81	0.56	14.22	2.83
	Static F1	14.8 _{2.2}	15.3 _{3.1}	14.8 _{2.5}	14.9 _{2.8}	13.6 _{2.6}	55.5 _{5.1}	21.5
	Adaptive F1	77.6 _{1.7}	76.5 _{2.3}	77.1 _{0.8}	77.7 _{1.0}	77.4 _{1.2}	88.7 _{1.1}	79.2
	Relative Gain	0.120	0.122	0.122	0.135	0.124	0.174_{1.2}	0.133
LLR	Utility F1	48.1 _{0.0}	48.1 _{0.0}	48.1 _{0.0}	48.1 _{0.0}	48.1 _{0.0}	58.3 _{17.7}	49.8
	CS	0.348	0.349	0.348	0.383	0.347	0.594	0.395
	Perplexity	693	668	693	674	689	314	622
	PI Masking	0.37	0.53	0.44	0.88	0.58	14.38	2.86
	Static F1	12.8 _{2.7}	12.4 _{2.8}	12.7 _{3.3}	13.3 _{2.9}	13.5 _{3.2}	53.7 _{5.8}	19.8
	Adaptive F1	75.7 _{2.2}	76.2 _{2.3}	76.5 _{2.0}	77.2 _{2.3}	75.6 _{4.6}	87.4 _{1.5}	78.1
	Relative Gain	0.133	0.133	0.131	0.143₂	0.130	0.088	0.126
t-score	Utility F1	48.1 _{0.0}	48.1 _{0.0}	48.1 _{0.0}	48.1 _{0.0}	48.1 _{0.0}	69.1 _{18.2}	51.6
	CS	0.362	0.363	0.362	0.396	0.362	0.591	0.406
	Perplexity	751	754	720	727	745	312	668
	PI Masking	0.45	0.55	0.54	0.86	0.59	14.38	2.90
	Static F1	12.2 _{2.0}	12.6 _{2.1}	12.1 _{2.8}	13.5 _{2.5}	12.7 _{2.0}	55.9 _{4.2}	19.8
	Adaptive F1	76.7 _{3.3}	77.1 _{1.7}	77.4 _{0.7}	77.6 _{1.4}	76.6 _{2.7}	88.0 _{1.7}	78.9
	Relative Gain	0.139	0.137	0.137	0.148_{1.2}	0.137	0.135	0.139
POS	Utility F1	48.1 _{0.0}	48.1 _{0.0}	48.1 _{0.0}	48.1 _{0.0}	48.1 _{0.0}	65.2 _{15.9}	50.9
	CS	0.452	0.453	0.454	0.472	0.453	0.712	0.499
	Perplexity	596	594	567	578	589	281	534
	PI Masking	4.18	4.34	4.30	4.56	4.33	15.59	6.22
	Static F1	18.2 _{3.9}	17.6 _{3.5}	17.2 _{3.4}	19.0 _{3.1}	18.2 _{3.4}	59.6 _{4.5}	24.9
	Adaptive F1	78.6 _{2.6}	81.6 _{0.4}	80.7 _{2.8}	80.5 _{0.5}	79.5 _{1.1}	89.0 _{1.4}	81.6
	Relative Gain	0.147₁	0.139₁	0.144₁	0.147	0.144₁	0.158₂	0.146
WordNet	Utility F1	48.1 _{0.0}	48.1 _{0.0}	48.1 _{0.0}	48.1 _{0.0}	48.1 _{0.0}	69.6 _{18.7}	51.6
	CS	0.323	0.324	0.324	0.358	0.323	0.593	0.374
	Perplexity	827	825	852	795	811	296	734
	PI Masking	0.55	0.72	0.72	1.07	0.81	15.70	3.26
	Static F1	15.1 _{4.7}	14.6 _{3.2}	14.3 _{3.8}	15.4 _{3.7}	14.4 _{3.4}	62.6 _{4.8}	22.7
	Adaptive F1	78.8 _{2.3}	80.3 _{1.6}	80.7 _{0.5}	79.0 _{0.4}	78.7 _{1.3}	90.4 _{0.2}	81.3
	Relative Gain	0.101	0.098	0.097	0.116₂	0.103	0.102	0.103
Average	Utility F1	48.1	48.1	48.1	48.1	48.1	67.8	—
	CS	0.367	0.368	0.367	0.398	0.367	0.616	—
	Perplexity	717	706	705	695	704	305	—
	PI Masking	1.18	1.32	1.31	1.64	1.37	14.85	—
	Static F1	14.6	14.5	14.2	15.2	14.5	57.5	—
	Adaptive F1	77.5	78.3	78.5	78.4	77.6	88.7	—
Relative Gain		0.128	0.126	0.126	0.138	0.128	0.131	—

Table 9: Results for the Yelp dataset at $\varepsilon = 18.7$. Each combination of decomposition and distribution strategy yields scores for Utility F1 ($\uparrow$), Cosine Similarity (CS, $\uparrow$), Perplexity ($\downarrow$), PI Masking ($\downarrow$), Static F1 ($\downarrow$), Adaptive F1 ($\downarrow$), and Relative Gain ($\uparrow$). Values derived from an average of three runs are displayed as mean_{std}. F1 values are in %. The rightmost column and bottom rows calculate the average results over decomposition and distribution methods, respectively. Best average Relative Gain values are **bolded**. For individual Relative Gain values, only the best are bolded: x_1 for best per decomposition, x_2 for best per budget distribution, and $x_{1,2}$ if both. The global best Relative Gain (across all combinations) is underlined.

Yelp $\varepsilon = 187$		Budget Distribution						Average
Decomposition	Baseline	Baseline	Attention	Gradients	IC	KeyBERT	YAKE	
		Utility: 87.8 _{1.9} / Adversary: 94.4 _{0.2}						
PMI	Utility F1	48.1 _{0.0}	48.1 _{0.0}	48.1 _{0.0}	48.1 _{0.0}	51.1 _{5.3}	85.3 _{2.6}	54.8
	CS	0.557	0.549	0.550	0.578	0.505	0.829	0.595
	Perplexity	484	486	455	463	480	144	419
	PI Masking	6.17	6.61	6.45	6.45	6.83	22.66	9.19
	Static F1	29.7 _{4.5}	28.5 _{5.9}	29.0 _{4.4}	30.8 _{5.5}	29.8 _{4.9}	81.9 _{1.8}	38.3
	Adaptive F1	80.9 _{0.8}	79.5 _{2.9}	80.8 _{1.6}	81.0 _{2.9}	79.4 _{0.5}	90.6 _{1.5}	82.0
	Relative Gain	0.148	0.152	0.146	0.154	0.139	0.220₂	0.160
LLR	Utility F1	48.1 _{0.0}	48.1 _{0.0}	48.1 _{0.0}	51.9 _{6.7}	48.1 _{0.0}	57.9 _{12.8}	50.3
	CS	0.549	0.544	0.543	0.572	0.499	0.601	0.551
	Perplexity	491	492	491	471	485	352	464
	PI Masking	5.31	5.98	5.84	6.06	6.37	7.92	6.25
	Static F1	28.2 _{5.4}	28.5 _{4.5}	29.5 _{5.0}	29.8 _{5.3}	29.6 _{5.2}	39.0 _{5.7}	30.8
	Adaptive F1	81.4 _{1.1}	80.8 _{2.4}	79.6 _{1.5}	80.0 _{3.0}	80.2 _{3.0}	83.6 _{1.1}	80.9
	Relative Gain	0.150	0.146	0.147	0.180₂	0.119	0.176	0.153
t-score	Utility F1	48.1 _{0.0}	48.1 _{0.0}	48.1 _{0.0}	48.1 _{0.0}	51.0 _{3.1}	83.2 _{2.1}	54.4
	CS	0.558	0.552	0.551	0.580	0.506	0.832	0.596
	Perplexity	479	480	480	459	474	142	419
	PI Masking	6.42	7.02	6.60	6.82	7.35	23.06	9.54
	Static F1	29.0 _{4.2}	28.9 _{5.2}	28.4 _{4.5}	30.8 _{5.4}	30.5 _{5.7}	82.5 _{1.2}	38.3
	Adaptive F1	82.0 _{0.9}	80.9 _{2.0}	82.2 _{2.6}	80.9 _{1.5}	81.8 _{2.2}	90.1 _{1.2}	83.0
	Relative Gain	0.147	0.145	0.143	0.154	0.127	0.209₂	0.154
POS	Utility F1	48.1 _{0.0}	53.1 _{8.7}	48.1 _{0.0}	55.2 _{8.4}	59.7 _{12.3}	84.8 _{0.8}	58.1
	CS	0.646	0.644	0.645	0.662	0.620	0.860	0.680
	Perplexity	504	504	500	484	489	159	440
	PI Masking	8.06	8.72	8.35	8.77	9.05	21.36	10.72
	Static F1	32.7 _{7.2}	33.3 _{7.7}	32.3 _{5.8}	33.7 _{7.6}	33.3 _{6.6}	78.2 _{1.8}	40.6
	Adaptive F1	80.0 _{1.9}	82.3 _{0.5}	83.5 _{1.2}	84.6 _{1.7}	82.3 _{1.5}	90.3 _{1.3}	83.8
	Relative Gain	0.182₁	0.195₁	0.169₁	0.207₁	0.217₁	0.252_{1,2}	0.204
WordNet	Utility F1	56.3 _{14.3}	48.1 _{0.0}	48.1 _{0.0}	57.7 _{16.8}	53.5 _{7.2}	86.5 _{0.5}	58.4
	CS	0.568	0.558	0.560	0.588	0.503	0.844	0.604
	Perplexity	521	523	494	496	520	130	447
	PI Masking	4.28	4.63	4.54	4.79	4.96	23.04	7.71
	Static F1	34.5 _{5.1}	32.7 _{6.6}	34.4 _{5.9}	35.9 _{5.5}	34.3 _{5.3}	85.3 _{0.8}	42.9
	Adaptive F1	82.6 _{2.1}	82.9 _{1.7}	84.6 _{1.9}	83.6 _{1.1}	82.3 _{2.5}	91.2 _{0.5}	84.6
	Relative Gain	0.182₁	0.137	0.126	0.190	0.132	0.219₂	0.164
Average	Utility F1	49.7	49.1	48.1	52.2	52.7	79.5	—
	CS	0.576	0.569	0.570	0.596	0.527	0.793	—
	Perplexity	495	497	484	475	490	185	—
	PI Masking	6.05	6.59	6.36	6.58	6.91	19.61	—
	Static F1	30.8	30.4	30.7	32.2	31.5	73.4	—
	Adaptive F1	81.4	81.3	82.2	82.0	81.2	89.2	—
	Relative Gain	0.162	0.155	0.147	0.177	0.147	0.215	—

Table 10: Results for the Yelp dataset at $\varepsilon = 187$. Each combination of decomposition and distribution strategy yields scores for Utility F1 ($\uparrow$), Cosine Similarity (CS, $\uparrow$), Perplexity ($\downarrow$), PI Masking ($\downarrow$), Static F1 ($\downarrow$), Adaptive F1 ($\downarrow$), and Relative Gain ($\uparrow$). Values derived from an average of three runs are displayed as mean_{std} . F1 values are in %. The rightmost column and bottom rows calculate the average results over decomposition and distribution methods, respectively. Best average Relative Gain values are **bolded**. For individual Relative Gain values, only the best are bolded: $\mathbf{x}_1$ for best per decomposition, $\mathbf{x}_2$ for best per budget distribution, and $\mathbf{x}_{1,2}$ if both. The global best Relative Gain (across all combinations) is underlined.

Yelp $\varepsilon = 935$		Budget Distribution						Average
Decomposition	Baseline	Baseline	Attention	Gradients	IC	KeyBERT	YAKE	
		Utility: 87.8 _{1.9} / Adversary: 94.4 _{0.2}						
PMI	Utility F1	84.9 _{1.7}	84.0 _{1.5}	83.0 _{2.0}	85.1 _{2.5}	85.1 _{2.1}	86.7 _{1.3}	84.8
	CS	0.887	0.874	0.873	0.883	0.796	0.962	0.879
	Perplexity	153	156	156	153	171	99	148
	PI Masking	21.10	21.30	20.83	21.36	21.85	28.31	22.46
	Static F1	74.8 _{1.8}	72.1 _{2.5}	73.5 _{2.5}	74.9 _{2.4}	72.0 _{1.9}	88.0 _{1.8}	75.9 _{0.0}
	Adaptive F1	88.5 _{1.6}	89.9 _{1.4}	88.2 _{0.9}	88.6 _{1.6}	90.1 _{0.6}	91.2 _{0.8}	89.4
	Relative Gain	0.286₂	0.278	0.275	0.283	0.240	0.256	0.270
LLR	Utility F1	86.0 _{0.3}	83.9 _{1.6}	86.1 _{1.3}	84.5 _{0.3}	84.5 _{1.5}	87.9 _{0.8}	85.5
	CS	0.881	0.868	0.868	0.878	0.788	0.962	0.874
	Perplexity	157	161	161	158	176	99	152
	PI Masking	20.34	20.45	20.20	20.64	20.78	28.35	21.79
	Static F1	74.1 _{1.9}	70.5 _{2.5}	72.1 _{2.5}	74.3 _{2.3}	70.8 _{3.1}	87.2 _{1.3}	74.8
	Adaptive F1	89.9 _{1.9}	88.8 _{2.1}	90.0 _{1.1}	88.7 _{1.4}	87.9 _{1.1}	91.8 _{1.2}	89.5
	Relative Gain	0.289₂	0.286	0.289₂	0.282	0.248	0.263₁	0.276
t-score	Utility F1	83.1 _{4.3}	82.5 _{1.8}	81.9 _{2.3}	83.4 _{1.8}	81.2 _{1.7}	84.3 _{1.8}	82.7
	CS	0.891	0.877	0.878	0.887	0.796	0.966	0.882
	Perplexity	151	154	154	152	169	98	146
	PI Masking	21.74	21.94	21.58	21.86	22.04	28.52	22.95
	Static F1	76.0 _{1.6}	72.9 _{2.3}	73.9 _{1.7}	76.2 _{2.1}	72.4 _{2.4}	87.6 _{1.8}	76.5
	Adaptive F1	89.2 _{1.0}	87.7 _{1.1}	89.0 _{1.8}	88.8 _{1.0}	89.3 _{1.2}	91.5 _{0.9}	89.2
	Relative Gain	0.270	0.275₂	0.265	0.269	0.220	0.245	0.257
POS	Utility F1	83.1 _{2.3}	83.4 _{1.1}	80.2 _{1.9}	80.9 _{1.8}	81.1 _{1.6}	84.5 _{1.7}	82.2
	CS	0.869	0.863	0.863	0.869	0.823	0.941	0.871
	Perplexity	198	199	199	197	207	122	187
	PI Masking	18.73	19.01	18.89	18.65	19.04	25.66	20.00
	Static F1	64.5 _{3.7}	63.4 _{3.5}	62.8 _{4.2}	64.9 _{4.2}	63.1 _{4.3}	84.1 _{2.0}	67.1
	Adaptive F1	87.8 _{1.1}	87.0 _{1.0}	88.6 _{0.5}	89.4 _{1.0}	88.2 _{0.5}	90.5 _{0.4}	88.6
	Relative Gain	0.313₁	0.317_{1,2}	0.297₁	0.295₁	0.280₁	0.258	0.293
WordNet	Utility F1	84.7 _{0.5}	83.1 _{1.6}	83.4 _{1.0}	83.7 _{2.2}	85.0 _{0.8}	86.8 _{1.7}	84.4
	CS	0.876	0.865	0.863	0.875	0.781	0.962	0.870
	Perplexity	159	162	162	159	177	96	152
	PI Masking	17.71	18.04	17.70	18.05	18.57	28.89	19.83
	Static F1	74.9 _{2.0}	72.5 _{2.5}	73.7 _{2.8}	74.9 _{2.0}	71.7 _{2.1}	88.4 _{0.8}	76.0
	Adaptive F1	88.8 _{0.9}	89.4 _{0.8}	88.5 _{1.2}	88.5 _{0.6}	89.0 _{1.0}	92.3 _{1.0}	89.4
	Relative Gain	0.290₂	0.280	0.281	0.283	0.248	0.249	0.272
Average	Utility F1	84.4	83.4	82.9	83.5	83.4	86.0	—
	CS	0.881	0.869	0.869	0.878	0.797	0.959	—
	Perplexity	164	166	167	164	180	103	—
	PI Masking	19.92	20.15	19.84	20.11	20.46	27.94	—
	Static F1	72.8	70.3	71.2	73.0	70.0	87.0	—
	Adaptive F1	88.8	88.6	88.9	88.8	88.9	91.4	—
	Relative Gain	0.289	0.287	0.281	0.283	0.247	0.254	—

Table 11: Results for the Yelp dataset at $\varepsilon = 935$. Each combination of decomposition and distribution strategy yields scores for Utility F1 ($\uparrow$), Cosine Similarity (CS, $\uparrow$), Perplexity ($\downarrow$), PI Masking ($\downarrow$), Static F1 ($\downarrow$), Adaptive F1 ($\downarrow$), and Relative Gain ($\uparrow$). Values derived from an average of three runs are displayed as mean_{std}. F1 values are in %. The rightmost column and bottom rows calculate the average results over decomposition and distribution methods, respectively. Best average Relative Gain values are **bolded**. For individual Relative Gain values, only the best are bolded: x_1 for best per decomposition, x_2 for best per budget distribution, and $x_{1,2}$ if both. The global best Relative Gain (across all combinations) is underlined.

Decomp.	Dist.	Great website, products and customer service!: They we great. Delivery was 7-7 but the text me a smaller window the night before so. Didn't have to stay in all day which was great and they called when they were ten mins away. One item was cracked but when I called up I didn't have to sit on hold or go through a complicated automated system. I got straight through to someone who arranged to send another item out ASAP. Really pleased
PMI	Baseline	anthem help create, and products or services!: they we nice. may still was career average - but the attorneys me a car club unable to move the penalty minutes before so. didn't have to gotten in all contact form which was fields and they attempt was made when they were research project canonical. great outdoors was inclusion criteria but when i list up i didn't have to organizers on seamless or i'll through a downloaded windows registry, i cursor close eye through to opened fire who nerdwallet to tenofovir sure everything out field. fun to make
	Attention	funeral home chapel million in revenue, and products and services!: they we sure that everyone. fields like was within 30 - but the conversation going me a penalty exclusive license the round pick before so. didn't have to halftime in all jel classification which was late 1800 and they annex when they were lieutenants identifiers. clothing company was would be perfect but when i deforestation up i didn't have to hurl on things going or google account through a runtime effortless, i really liked second goal through to minaj who would fit to stipulated big bang theory out available on itunes. harshly
	Gradients	relegated family style, and higher revenue!: they we snow. started writing was new poll - but the tracking number me a wagering requirements primary research the looked amazing before so. didn't have to feasibility study in all scheduled for release which was settings page and they consisting when they were pastures time to put. html file was awarded but when i cram up i didn't have to conversation going on regions or choose us through a cute extensive range, i bought jangle through to sure everyone who encrypted to shop childrens work sure everyone out wondering if anyone. really stood
	IC	would make a great wipo case, and available online!: they we footprint. questionnaire was 2 2 - but the fungus me a 4gb ever worked the vibrate before so. didn't have to frequently become in all quest which was minimalism and they mansion when they were tibetan min action adventure. default was hip hop but when i paved the way up i didn't have to start the day on anxiety or fun playing through a other streak, i pm powered by vbulletin sarah jessica parker through to fits well who seater to whole or in part offer a full out leach. pleaded guilty to one
	KEYBERT	momentum going get organized, and screenshots!: they we babip. online survey was nov - but the easily me a rapidly pokemon the action before so. didn't have to eventually become in all quest which was minimalism and they mansion when they were tibetan min action adventure. default was hip hop but when i paved the way up i didn't have to start the day on anxiety or fun playing through a other streak, i pm powered by vbulletin sarah jessica parker through to fits well who seater to whole or in part offer a full out leach. pleaded guilty to one
	YAKE	included studies mail buy, and client service!: they we editors. craig1916 was 7 7 - but the tolstoy me a long and short samsung galaxy the side before so. didn't have to clock ticking in all dresses which was presided and they localhost when they were terre were apparent attempt. purpose flour I was features built but when i decided to use up i didn't have to sit on hold or go through a complicated automated system, i got straight through to someone who arranged to send another 2019 03 out root access. username and password
LLR	Baseline	ungodly website, anti theft system google announced!: they we p m cst. ll was consumers may - 120hz but the apes me a helping the anyplace before so. didn't have to yet received in all awarded which was please and they peacekeeping operations when they were franchise history best outdoor. genetic counseling was meet the requirements but when i imprisoned up i didn't have to sitting on term life insurance or account with us through a complete stop story was originally, i second attempt pollen through to sore muscles who pushbutton to give rise served as the director out domain name. maceration
	Attention	software program zip file, data cannot insure!: they we conservator. finances was dating app - stocking but the akc me a full frame metacritic the feel before so. didn't have to take care in all took us which was member of the board and they let us when they were second period keyless entry. maps was co v but when i please feel free up i didn't have to weight room on best practices or grow through a fermentation ecommerce platform, i crying electric shock through to girl who knew i wanted to key fob food stalls out dedicated to helping. rear cross traffic
	Gradients	please payback period, across the web calculate!: they we winnings. yet released was 1xbet - push button but the termed me a isdn automatic the plate appearances before so. didn't have to forefoot in all motherboard which was final whistle and they justice when they were sarajevo nest. outbreak of war was parked but when i burned up i didn't have to straighten on video formats or reels through a recently received crypto currency, i panoramic sunroof swee through to remi maintenance fee reminder who wanted to see to might make drumsticks out cannot guarantee. made me realize
	IC	screen canva, last moment line casino!: they we it'd. win one was annual production - little did i know but the readmissions me a rooms are equipped several attempts the ajiga before so. didn't have to left to play in all morning which was mocking and they fascinating when they were professor nair. elantra was two day but when i retarded up i didn't have to sunday april on engineering manager or feel free through a struggling amount of water, i marries lame through to water flow who punished to actual damages another vehicle out six figures. year in a row
	KEYBERT	tailgate please feel free, last summer unsuccessful attempt!: they we mobs. yet received was seemed like - yet reached but the remote keyless entry me a situated eliminate the 970 before so. didn't have to international airports in all discipleship which was fun and they attempts when they were received run of the mill. odoo was appoint but when i didn't have to it possible up i didn't have to yell on take or get to go through a compares type of treatment, i previous clients 100 0 through to something who better person to please feel free gmo free out pills. purposed
	YAKE	if time and cost, strange at first data protection officer!: they we plugin. distributed was 7 - 7 but the independent contractor me a opting first try the friday night before so. didn't have to stay in all week since which was fun and they mm thick when they were g dl author david. discounted was clunky but when i lean startup up i didn't have to sit on hold or go through a complicated automated system, i got straight through to someone who arranged to send cares act out disappearance. really pleased
t-score	Baseline	term contract, 13mp!: they we bright future. pdt was gulps - but the median follow me a continue to grow materials the last days before so. didn't have to sects in all success which was always a good and they common cause when they were curse words get inspiration. construction debris was principalities but when i incubator up i didn't have to lay their eggs on gently or heart through a nests commercial drivers, i two girls transfer rate through to nugget who graze to large mixing overlap out academically. hard to get
	Attention	brief conversation, free sites!: they we caffe. forms including was 67 - but the selections me a topics including feet of water the world before so. didn't have to christmas in all greasy which was something and they uk s leading when they were reservations with confidence days to go. certificates of deposit was episode but when i recently asked up i didn't have to laugh on completed within or going to take through a hard to come time mixing, i feet high material including through to loved one who miles southwest to heavenly host prior written consent out happy life. bringing people together
	Gradients	pooja, bombay!: they we gamescom. reforms was episode - but the christmas special me a hotels honourable the closing before so. didn't have to resins in all eighth inning which was crates and they area known when they were hard to take players would also. sturdiness was barry bonds but when i elapsed up i didn't have to something on find many or spaced through a full potential early to say, i levees villain through to amicable who 1926 to excavation future growth out dad. jla
	IC	sprout, mobility issues!: they we yearbook. christmas was kilometers west - but the page numbers me a hamilton beach centre the rishikesh before so. didn't have to new episode in all entire life which was many products and they could live when they were aired fastest growing companies. every name was oozed but when i quae up i didn't have to moment on mpeg or think people through a hundred feet full potential, i teammates fairway through to sources including who plastered to inquiring items including out arthritic. therefore essential
	KEYBERT	balanced approach, characters including!: they we work closely. reputation was supporter - but the items including me a much more fun asbestos removal the sinai before so. didn't have to easier to see in all memories which was materials like and they uttered when they were please mention this item 11 00 11. valid reason was guru but when i household name up i didn't have to better to go on genesis 1 or buy through a intricate copyrights, i selection of games upstarts through to child who recipe calls to speak to us competencies out hard cock. new team
	YAKE	blog article, cmx!: they we yards and a touchdown. products including was 7 7 - but the text me a traverse oiling the cherry trees before so. didn't have to area near in all pagoda which was materials including and they blues when they were dispersed claude monet. prostitutes was terminator but when i application number us up i didn't have to sit on hold or go through a complicated automated system, i got straight through to someone who arranged to send another ebay out us to share. aspects of the game
POS	Baseline	great website products, and customer service!: they we 2019 20. was kelvin - but the link back me a fir the wick before so. didn't have to there's in all baking which was fragile and they tossed when they were network traffic appropriate action. medicine cabinet was free trade agreement but when i instrumental up i didn't have to sprung on crossed or exists through a complicated automated system, i angrily through to candidate who first paragraph spill out listeners. special issue first product
	Attention	great website products, and washing!: they we new file. was 7 4 - but the valid license me a planning to put the barking before so. didn't have to skin feels in all entire month which was cold front and they first full season when they were good fortune stairs. jennifer aniston was humidity but when i opened up i didn't have to sitting on convey or one god through a complicated automated system, i insults through to reasonable steps who designed to attract funny story out educators. quran indian institute
	Gradients	great website products, and leadership team!: they we slinging. was reigning - but the dug me a moment the supervise before so. didn't have to diocese in all basic needs which was pumped and they motorists when they were insects away. health reasons was carved but when i colonial up i didn't have to multiple choice questions on blood sample or raged through a complicated automated system, i harried through to reasonable efforts who entire list drab out grew. simple easy secretary
	IC	great website products, and unison!: they we documents related. was online sources - but the christmas morning me a expansion the old one before so. didn't have to relevant references in all nfl draft which was cannot rely and they global trade when they were trampling treadmill. picasso was lot bigger but when i buddhism up i didn't have to body composition on also receive or put through a complicated automated system, i decided to drive through to law who tiptoe fluorine out tallied. slowed loving god
	KEYBERT	great website products, and protests!: they we marshal. was oppo - but the legal obligation me a failed to act the sex offenders before so. didn't have to drying time in all bottle which was rich and they ate when they were grasses optioned. two glasses was offside but when i attacked up i didn't have to greet on unconscious mind or stumbles through a complicated automated system, i ask to speak through to one song who gross margin similar product out cooperation. never use happy
	YAKE	great website products, and customer service!: they we international federation. was 7 7 - but the text me a search keyword the night before so. didn't have to working directly in all massacred which was slowdown and they jammed when they were healthy gums away. someone was threw but when i last second up i didn't have to sit on hold or go through a complicated automated system, i got straight through to someone who arranged to send live and work out would taste. really commercialize
WordNet	Baseline	loopholes everest, prizes and subscribe service!: they we lacking. bingo was travelled - sleeps but the odometer me a fiscal years number the evening before so. didn't have to spins in all 10 which was snooze and they there'll when they were jazz lunch montgomery. trial shipment was dewy but when i focuses up i didn't have to poker games on shock waves or fiscal years through a objected alarming chassis, i finished antennae through to snooze who lacking to remind they'll securities and exchange commission out phys. very fiscal years
	Attention	bomber take advantage, 1xbet and inefficiencies insured!: they we smoother. raimi was 24 - 256 but the paylines me a perpetrate lacking the fiscal years before so. didn't have to keep in all months which was perfect and they undermines when they were rockets wonderful fibers. regiment solution was poked but when i wanna up i didn't have to expressway on gambling or coppers through a home buyer mission impossible infantry, i hung roof through to them who next door to donate accusation financial year out melinda. prove truth
	Gradients	yosemite who'll, raimi and brewery lenders!: they we fantastic. locomotives was can - trials but the townland me a there italian renaissance the nightclub before so. didn't have to stay in all preview which was wysiwyg and they tyne when they were decreased pokies unmoved. ecole electrolyte was bet but when i eerie up i didn't have to sit on announced or kings through a sharper iphone bonuses, i gets got through to truth who trial to confirm latest recipe out gaming. snooze shockwaves
	IC	manicured warship, products and 4in wi!: they we achieved. dont was 16 - 11 but the covariates me a sensitive fy the uncle before so. didn't have to reunite in all rows which was sparked and they fallacy when they were constantinople min loaded. filling reels was eraser but when i shattered up i didn't have to sit on slots or ginseng through a challenges proofing frequently, i lived onboard through to difficulty who aimed to send feige slots out gaming. revengeance msci
	KEYBERT	cope federal trade commission, games and breaks birth control!: they we humanistic. million was 00 - teaches but the experiments me a rocky usdfor the night before so. didn't have to tread in all regiment which was good time and they holy week when they were 170 feel pleasant. lacking calendar year was we'll but when i oppose up i didn't have to sit on wait or send through a possible let's secures, i fades water tanks through to sports car who wronged to respond impermanence construed out sto. drug addict misdiagnosed
	YAKE	1xbet website, items and aftermarket service!: they we lot. mailbox was 7 - 7 but the manuela me a sporty window the weeknight before so. didn't have to traverse city in all norte which was corsets and they adjudicating when they were five mins away. one thermometer was cracked but when i yelling up i didn't have to sit on hold or go through a complicated automated system, i got straight through to someone who arranged to send another indictment out they'll. really pleased

Table 12: Obfuscated output examples from the Trustpilot dataset, at a document-level $\varepsilon = 52$.